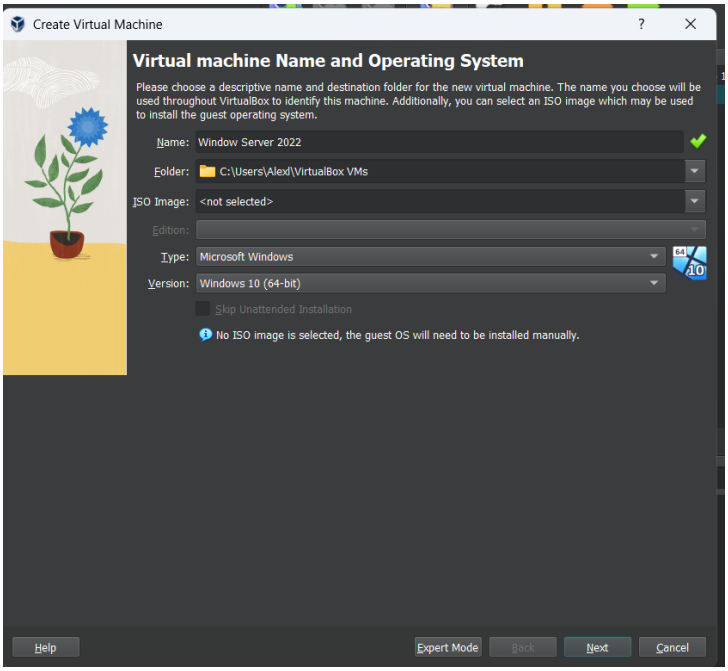
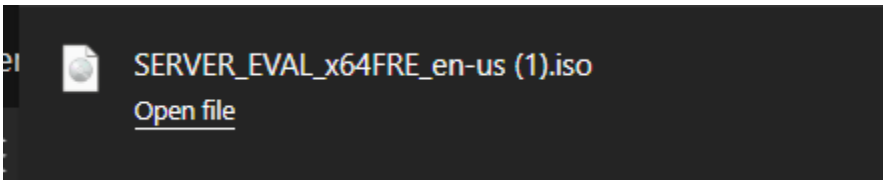
Alex Lang IT Helpdesk Home Lab

What I did during each day of this project

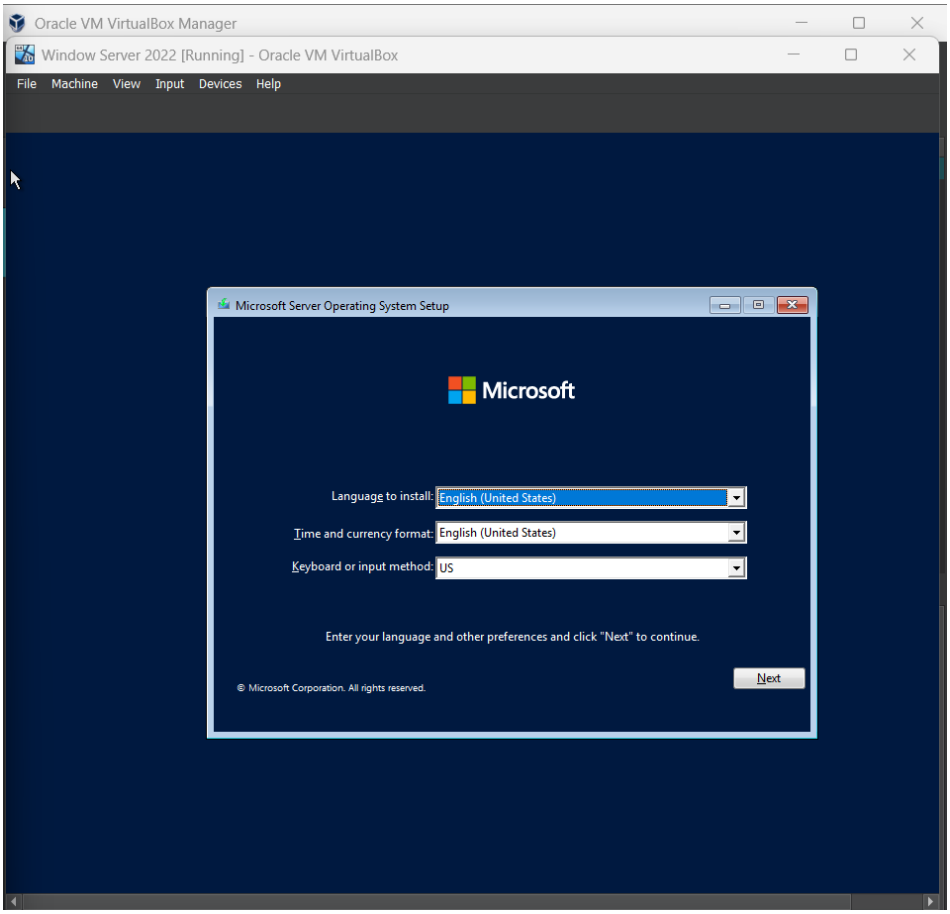
Day 1 Getting the VM Setup



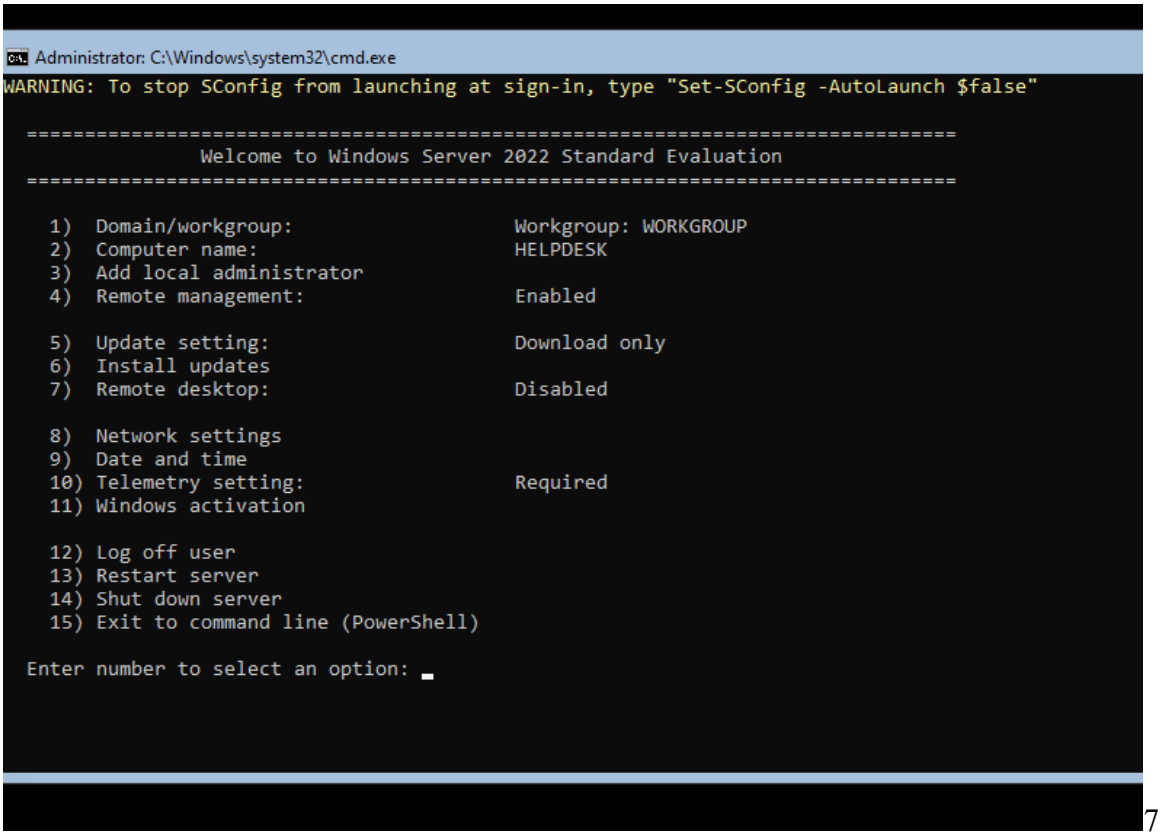
Created a new VM using VirtualBox to host the windows server



Downloaded the iso image of the server from microsofts website

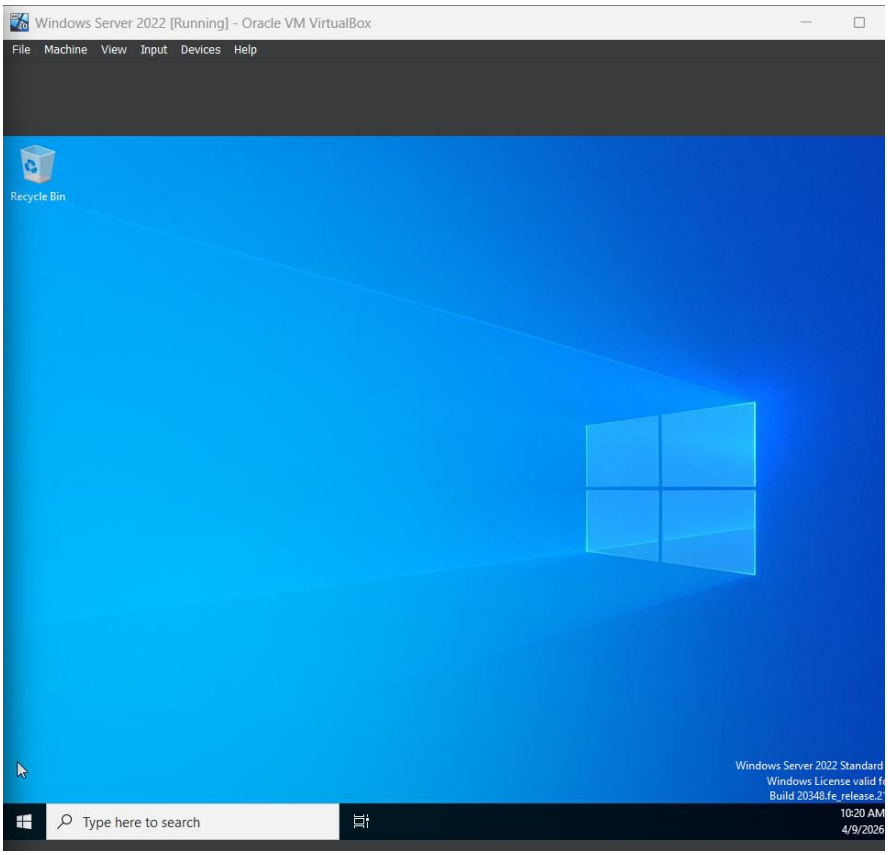


Started the VM, and after the boot failed mounted/booted the ISO file

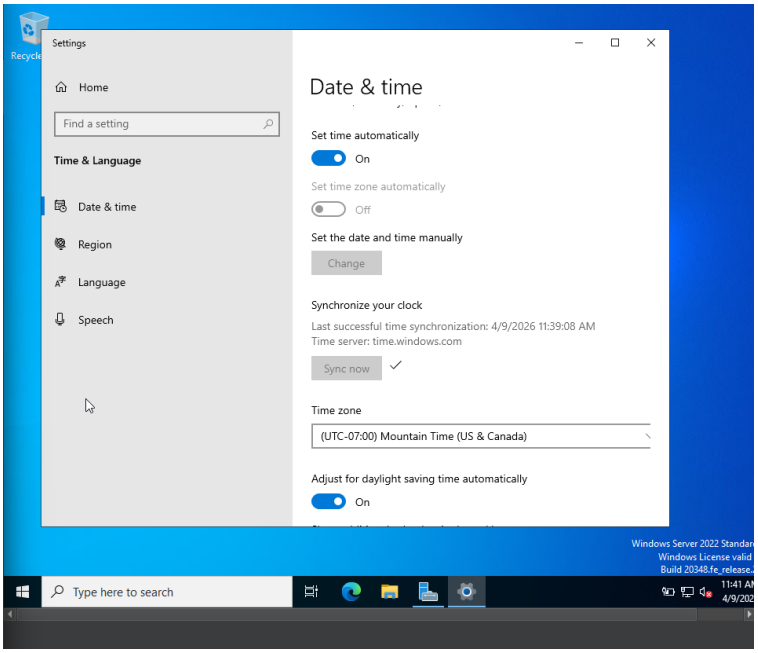


7

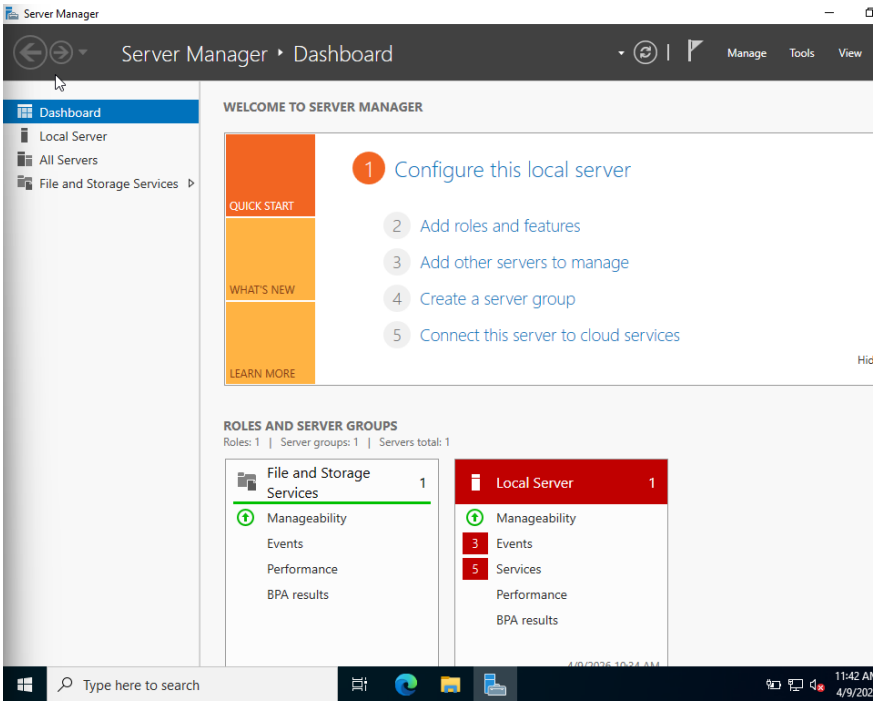
Changed some of the prebootup settings, specifically changed the computer name, enabled RDP, and changed the time settings as it was an hour behind Mountain time.



First time inside of the freshly booted VM



Changed time settings to sync up with my actual time zone

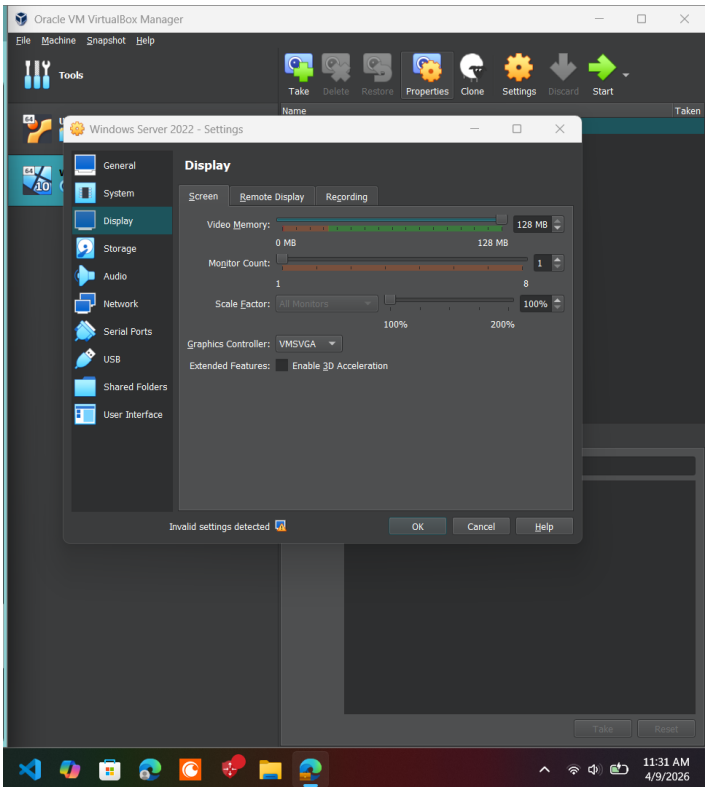


Server preconfigured

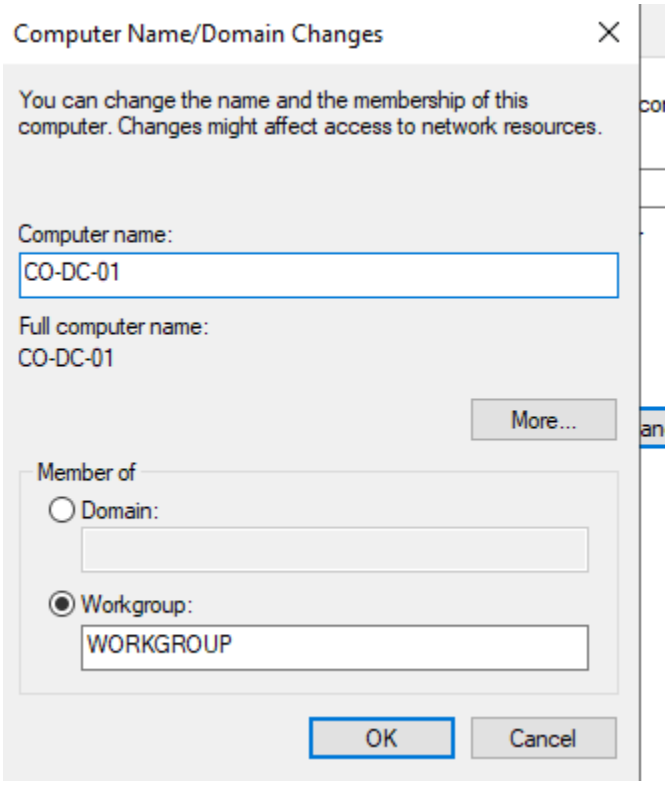
Finished the OS Setup, and left pretty much everything default as I can go back and edit later if needed

Observations/Problems that I had to troubleshoot through

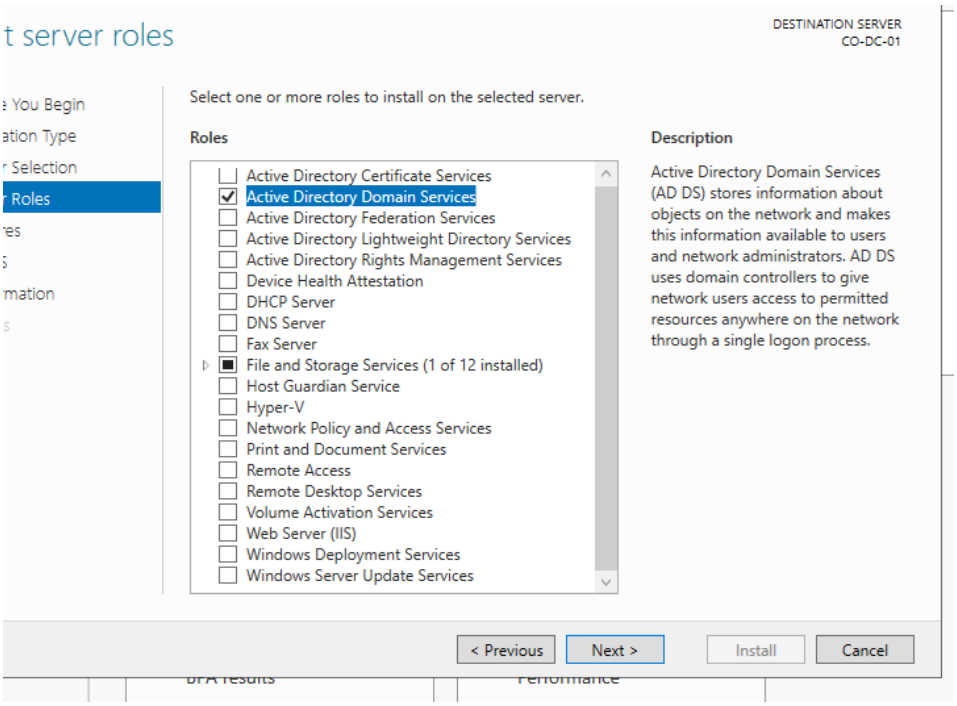
- When setting up the server, I didn't select desktop experience, so I had to redo the installation and select desktop experience(There might of been a way to just redo the installation but I thought it would be easier)
- Had some issues with the VM when initally setup where the screen would flash black and crashed, so I changed the graphics controller to VMVGA instead which fixed that prolbem



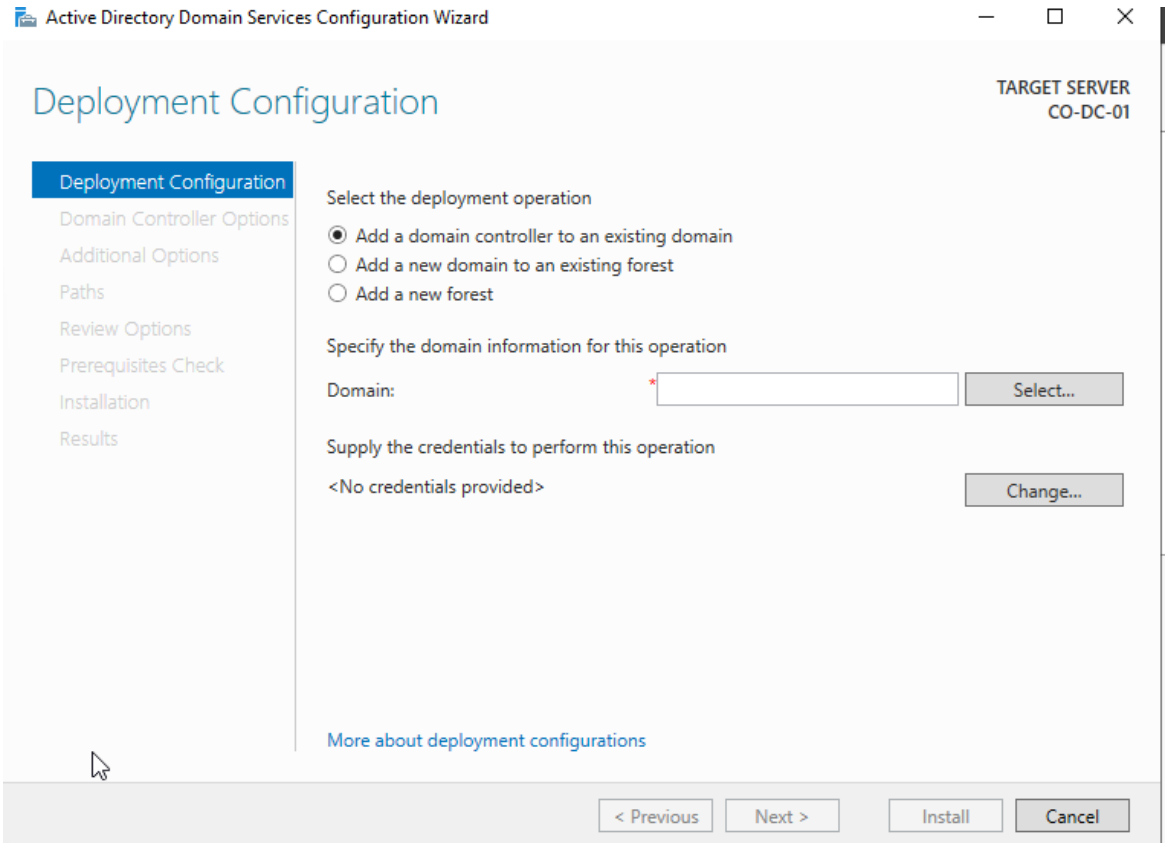
Day 2 Configuring/Installing AD



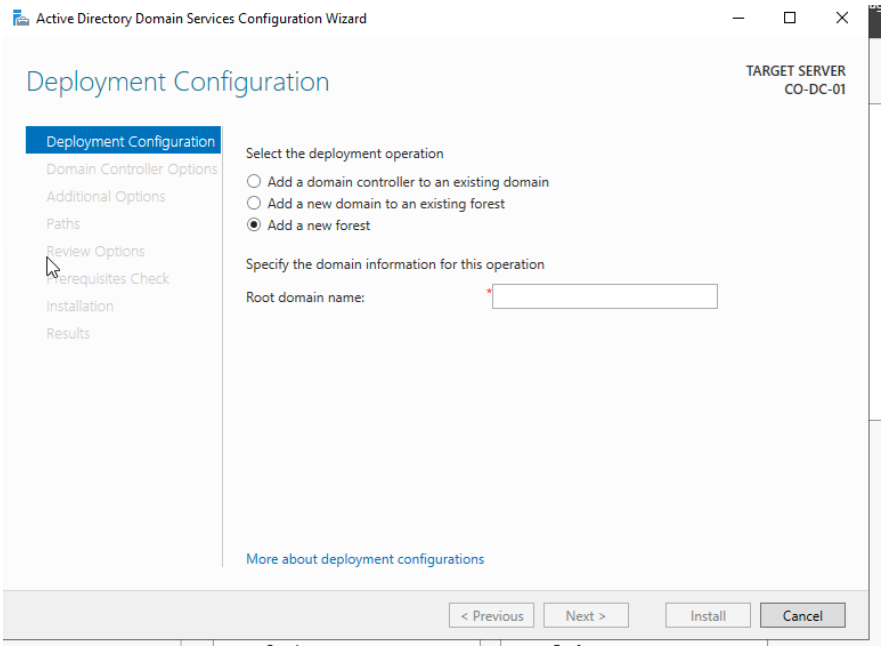
Changing computer/server name to CO-DC-01 to represent a real world domain server



Installing AD domain services



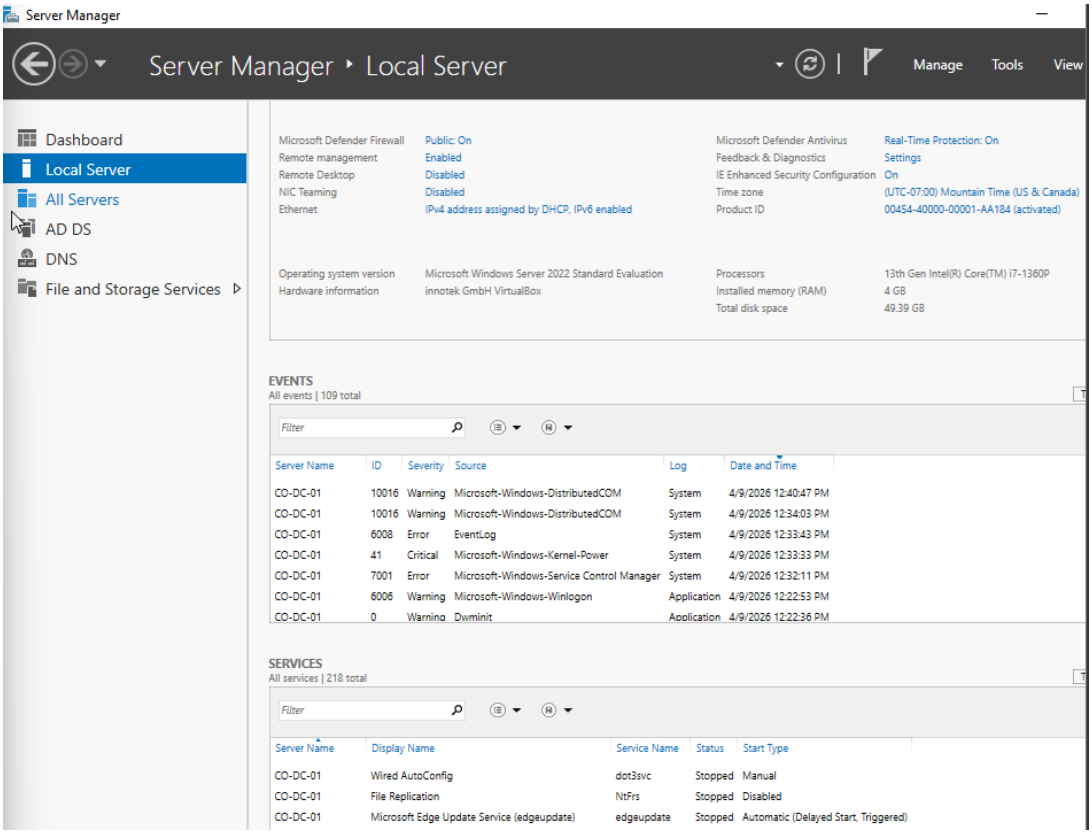
Promoting this server to a domain controller



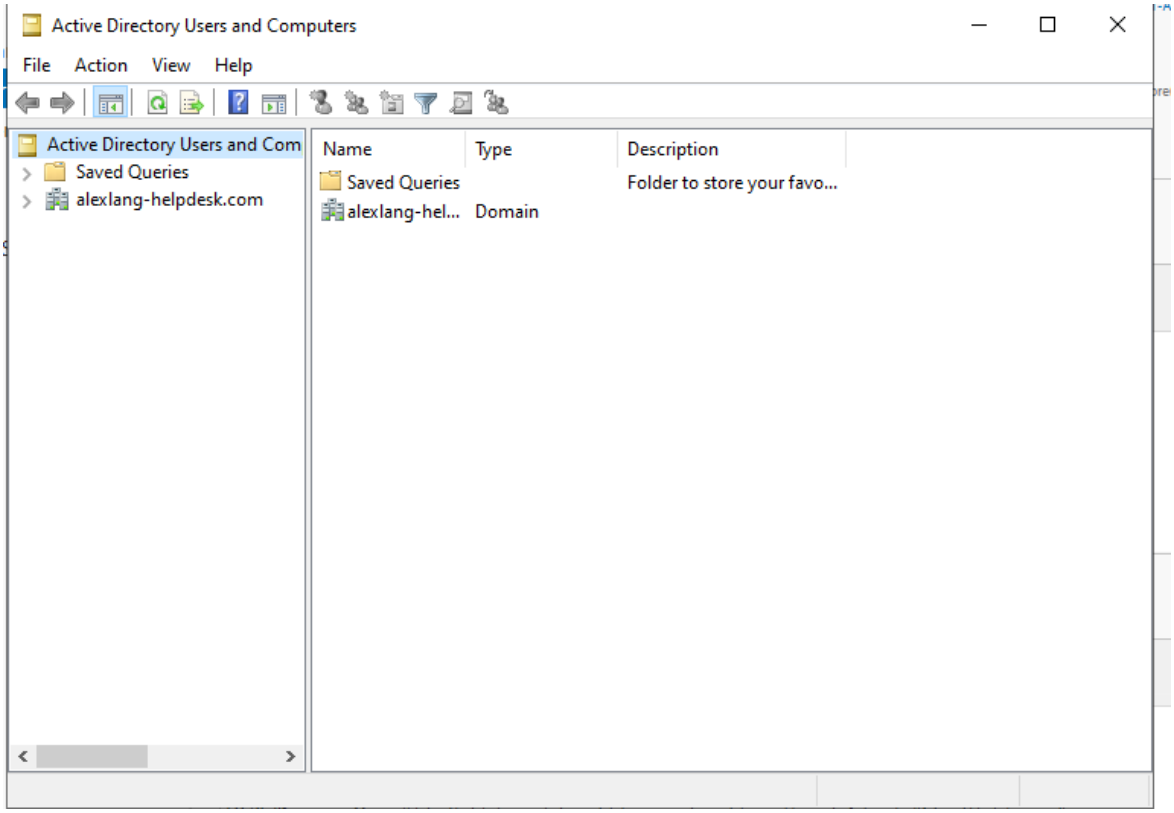
Need to configure as a new forest since there are no domains established yet

```
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>whoami
alexlang-helpde\administrator

C:\Users\Administrator>net use
New connections will be remembered.

There are no entries in the list.


C:\Users\Administrator>net accounts
Force user logoff how long after time expires?:       Never
Minimum password age (days):                          1
Maximum password age (days):                          42
Minimum password length:                              7
Length of password history maintained:                24
Lockout threshold:                                    Never
Lockout duration (minutes):                           30
Lockout observation window (minutes):                 30
Computer role:                                        PRIMARY
The command completed successfully.


C:\Users\Administrator>net user administator /domain
The user name could not be found.

More help is available by typing NET HELPMSG 2221.
```

Played around with some of the commands I learned from A+ including whoami, net user, net use, and systeminfo.exe



Domain successfully setup



Just ensuring that its setup, no users yet but will add them soon

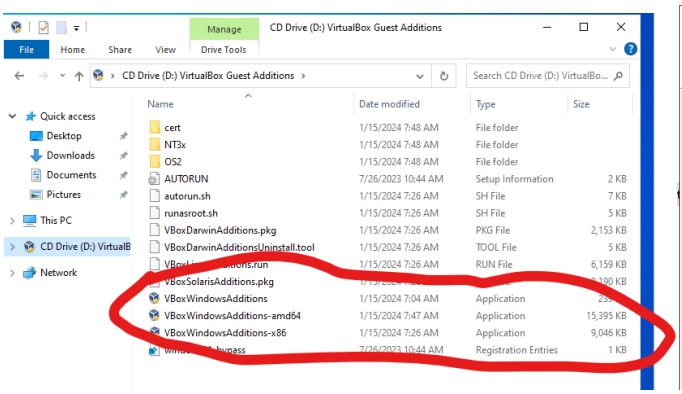
Observations/Problems that I had to troubleshoot through

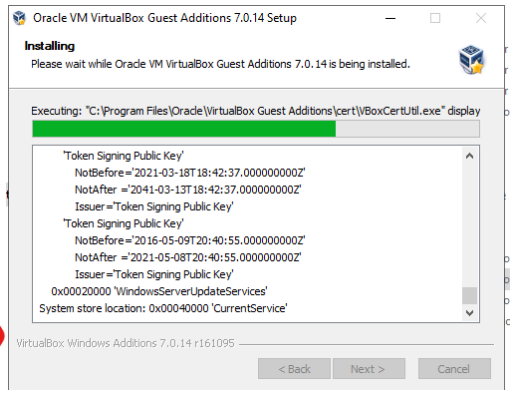
- The installation process of AD took much longer then I expected, but the VM doesn't have the highest amount of memory so it does make sense
- I noticed that once the AD was setup, there was over 100 security warnings(some critial, some less serious) which suprised me becuase I just assumed it would be slightly less vunerable

Server Name	ID	Severity	Source	Log	Date and Time
CO-DC-01	10016	Warning	Microsoft-Windows-DistributedCOM	System	4/9/2026 12:40:47 PM
CO-DC-01	10016	Warning	Microsoft-Windows-DistributedCOM	System	4/9/2026 12:34:03 PM
CO-DC-01	6008	Error	EventLog	System	4/9/2026 12:33:43 PM
CO-DC-01	41	Critical	Microsoft-Windows-Kernel-Power	System	4/9/2026 12:33:33 PM
CO-DC-01	7001	Error	Microsoft-Windows-Service Control Manager	System	4/9/2026 12:32:11 PM
CO-DC-01	6006	Warning	Microsoft-Windows-Winlogon	Application	4/9/2026 12:22:53 PM
CO-DC-01	0	Warning	Dwminit	Application	4/9/2026 12:22:36 PM

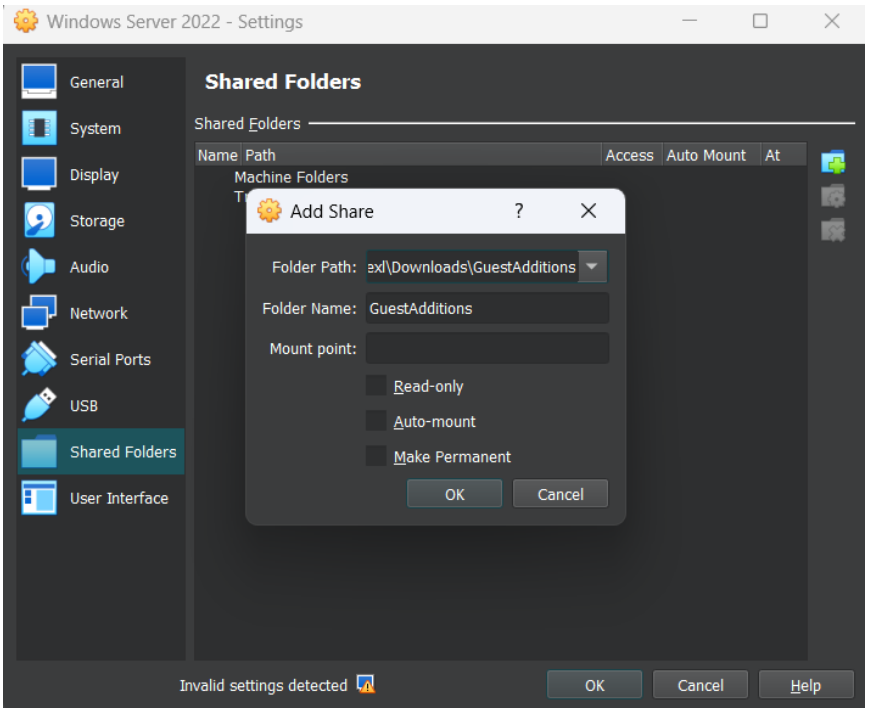
- Other than that not many issues

Day 3 Guest Additions / CD

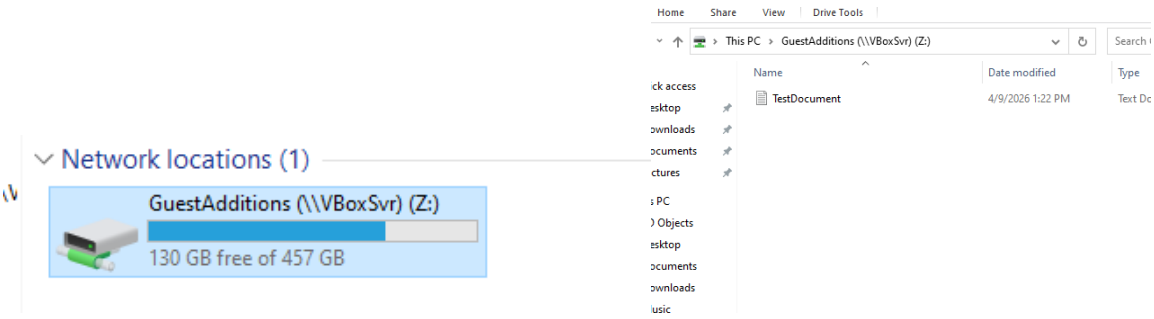




Found VboxWindowsAdditions/Installations on the C drive which I installed



Created a Guest folder on my home os that will be shared to the VM and used with Guest additions

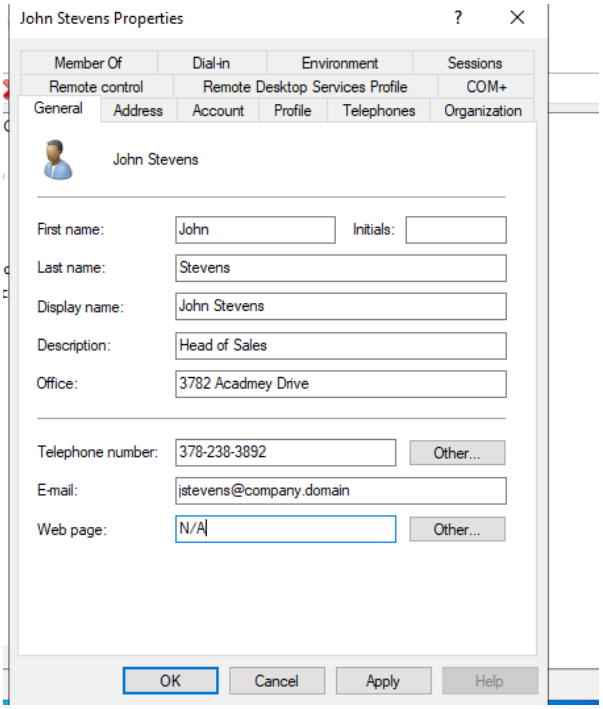


Guest addictions successfully configured and the test file was added.

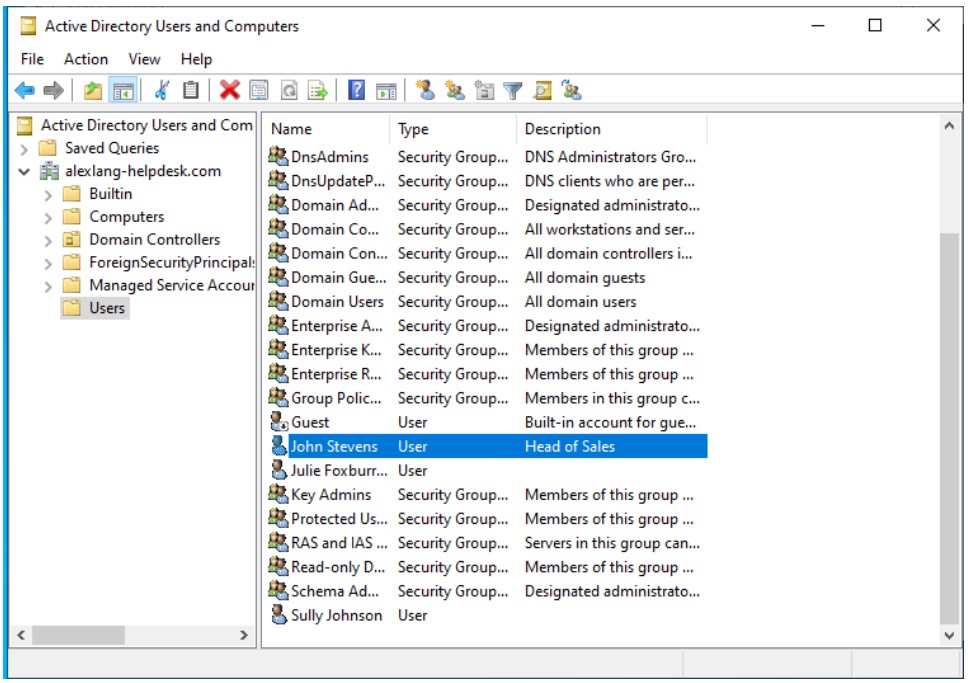
Observations/Problems that I had to troubleshoot through

- No issues, pretty straight forward day

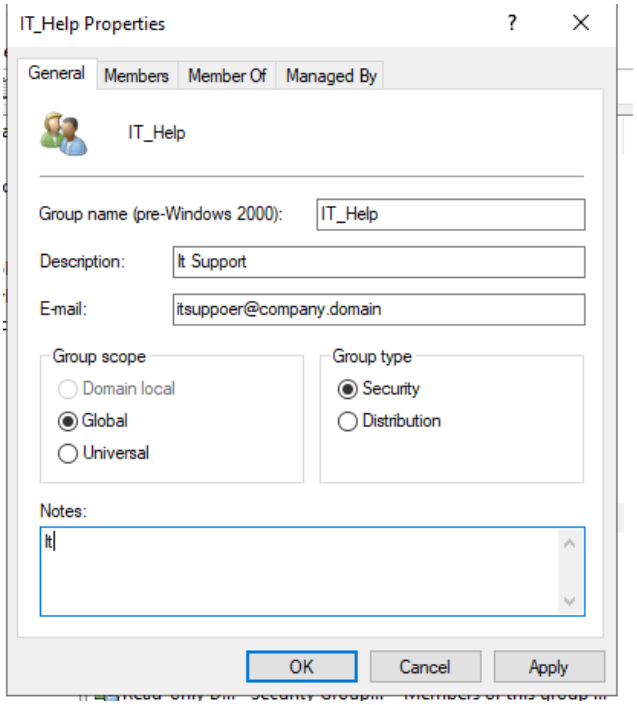
Day 4 Creating Users inside of AD



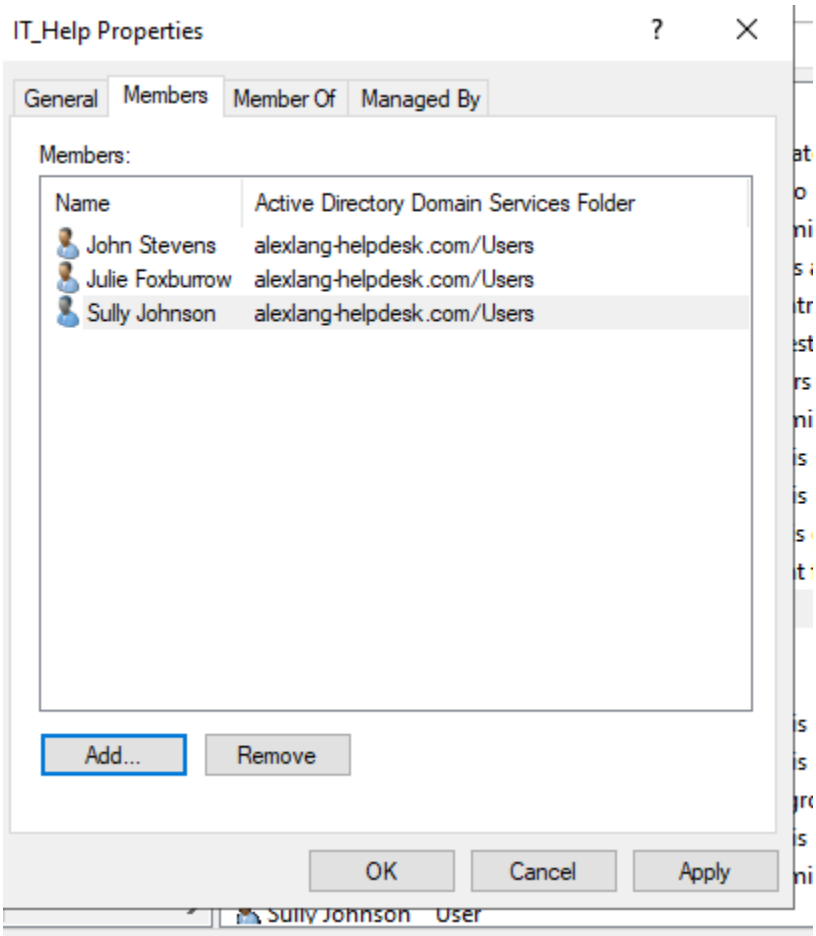
Created first admin user, and played around with the different settings/privileges



Created 2 more users so I can implement group policies

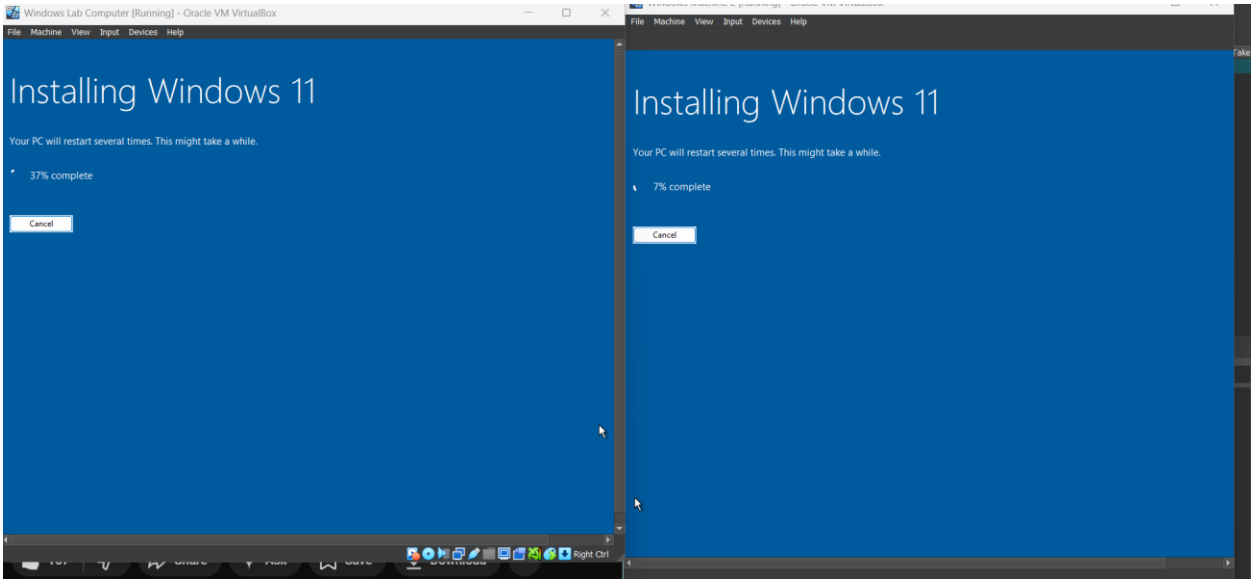


Createad new security group named IT_Help and added the 3 users just created to it

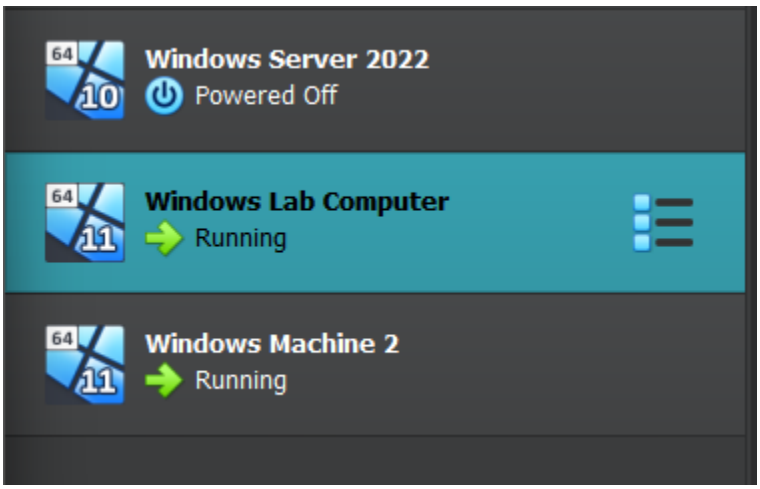


All users being added

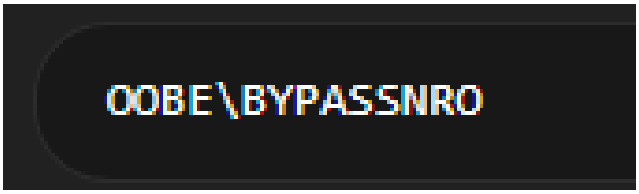
Day 5 Configuring a Windows 11 machines



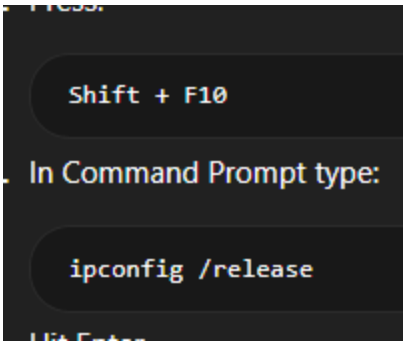
Installing 2 machines at once so I only need to wait for them to install once



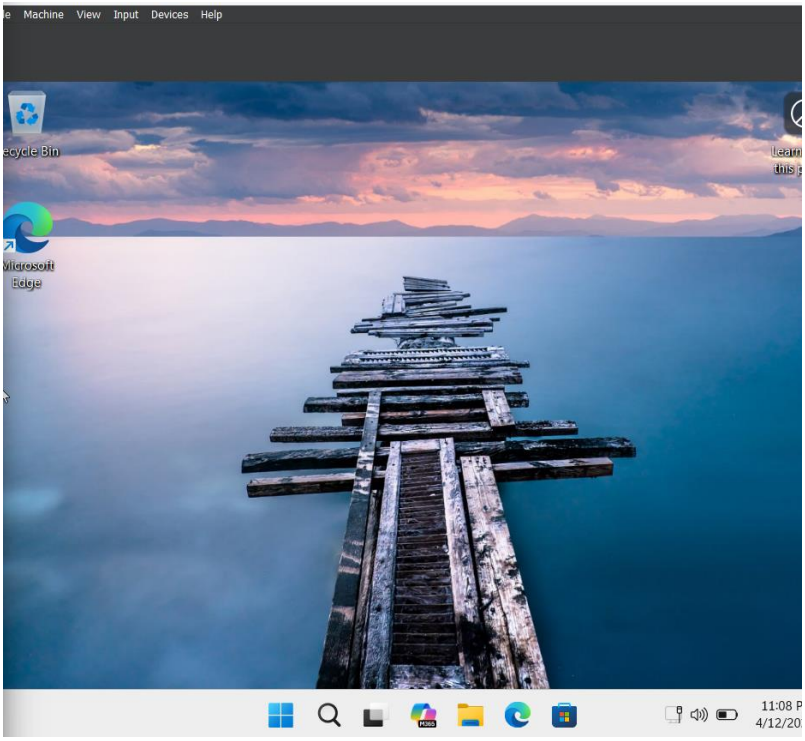
Now I have the sever and 2 machines that I can join to the domain created on the windows server



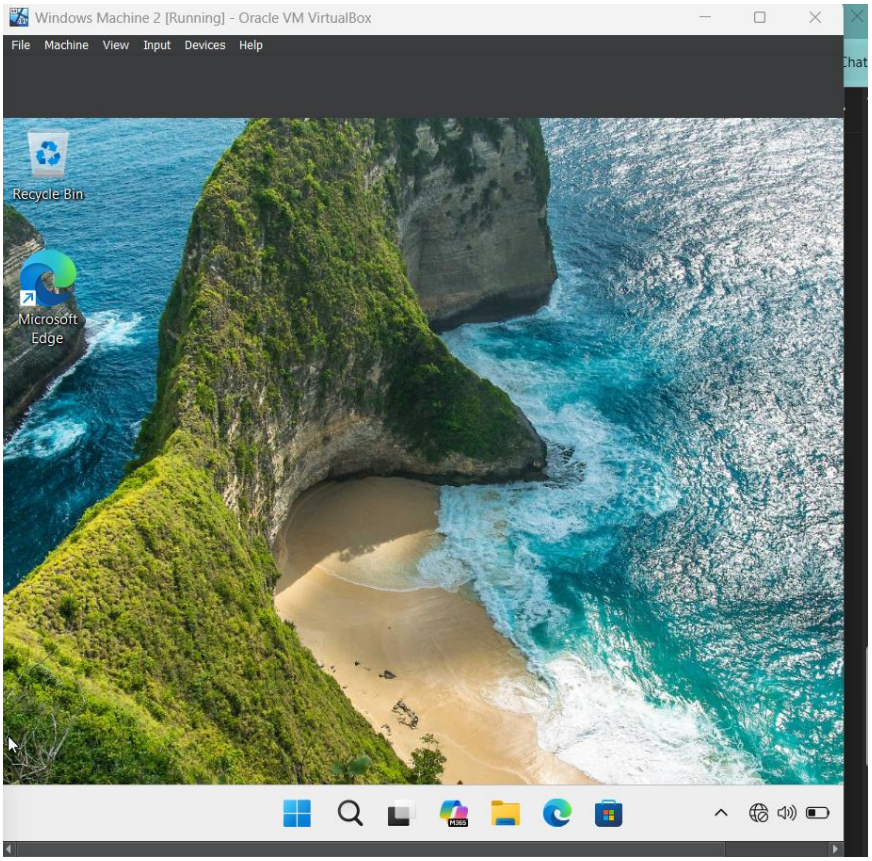
This command turned off the need to create an account, and bypassed it



After running the previous command, the OS got stuck in a checking for updates loop, so I had to release the connection for now



First workstation successfully setup

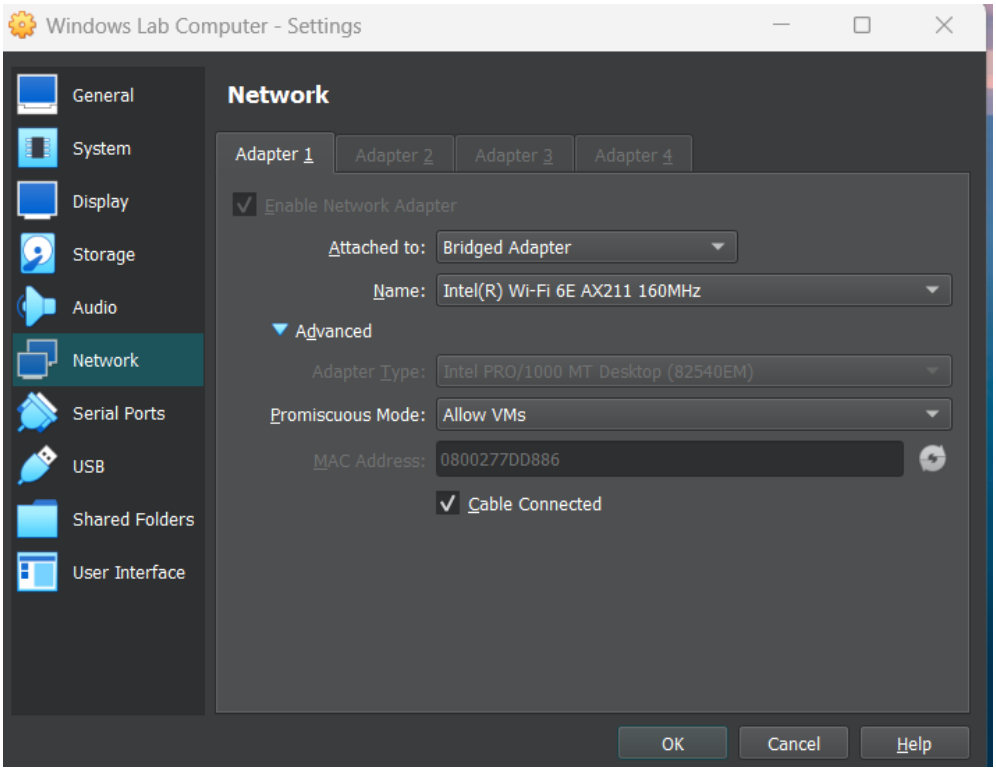


Second PC also good to go

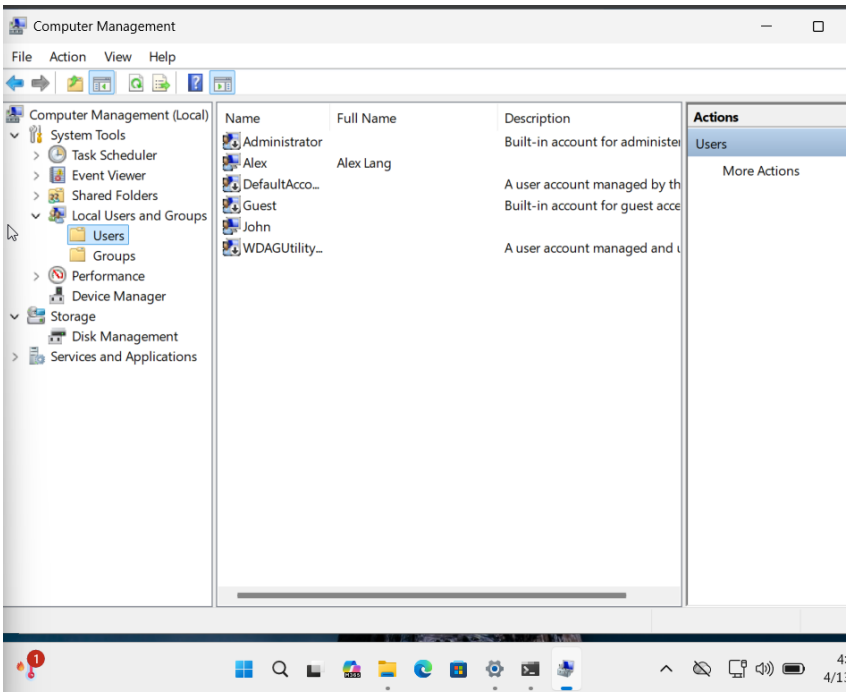
Observations/Problems that I had to troubleshoot through

- The Windows 11 ISO took a very long time to install
- I had an issue first booting the VM, but did some troubleshooting and pressing on the black screen and pressing a key helped
- The Windows 11 took a long time to install
- While trying to setup the PCs, they were trying to force me to make an account which I didn't want to do since this is just for experience practices(So I used AI to find a command to help me bypass this setting)
- The second PC had a different issue and did not want to boot so I had to go into the regedit and edit setup.exe to fix that error

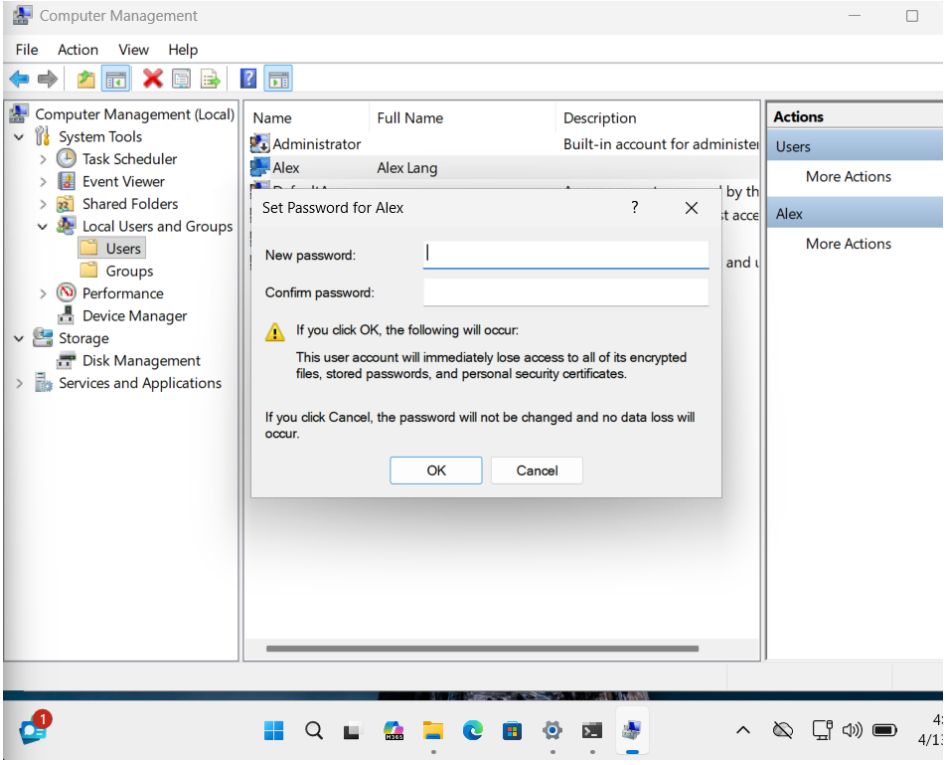
Day 6 Setting up the Domain Connection



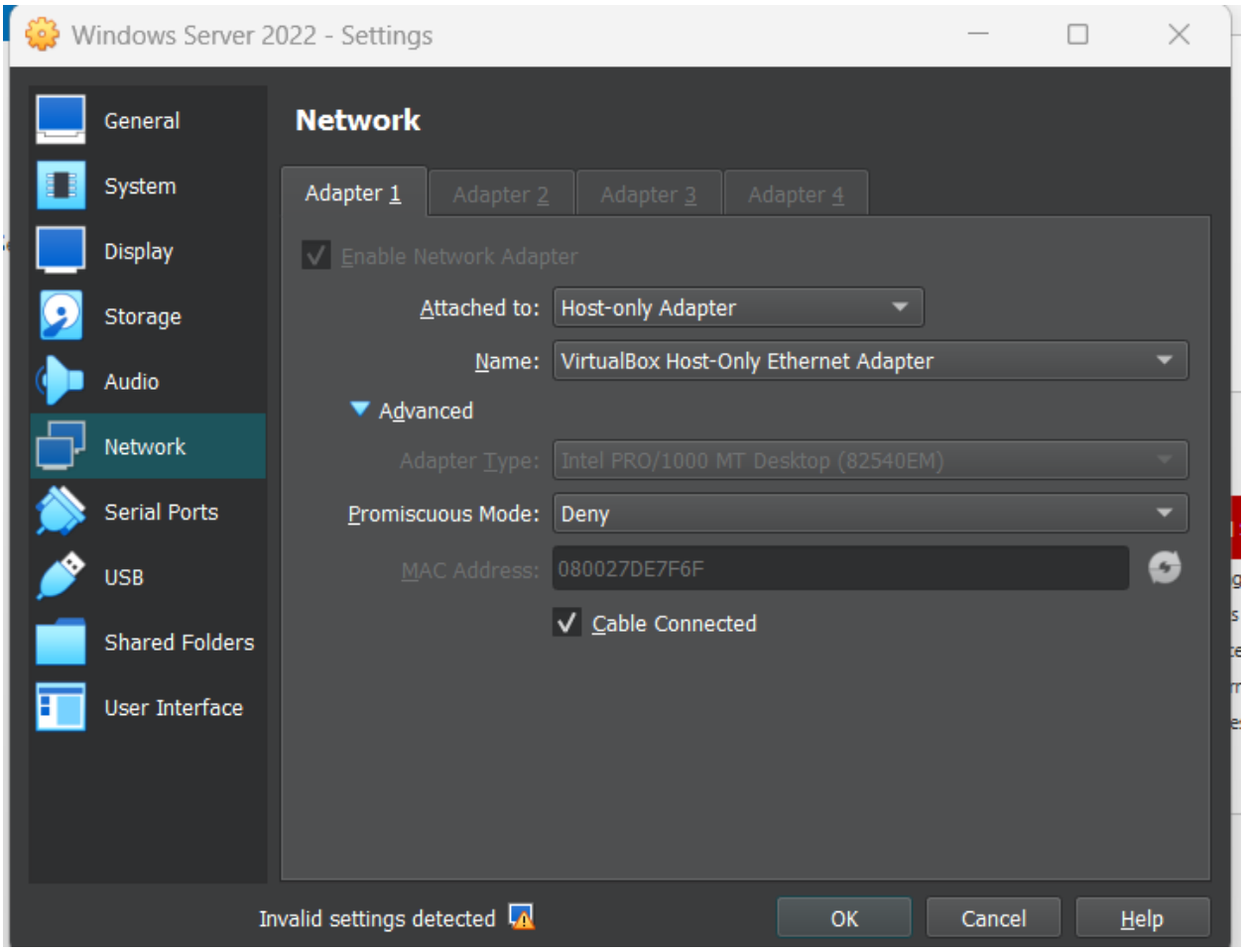
Updating Netwrok VM settings to allow VMs to connect



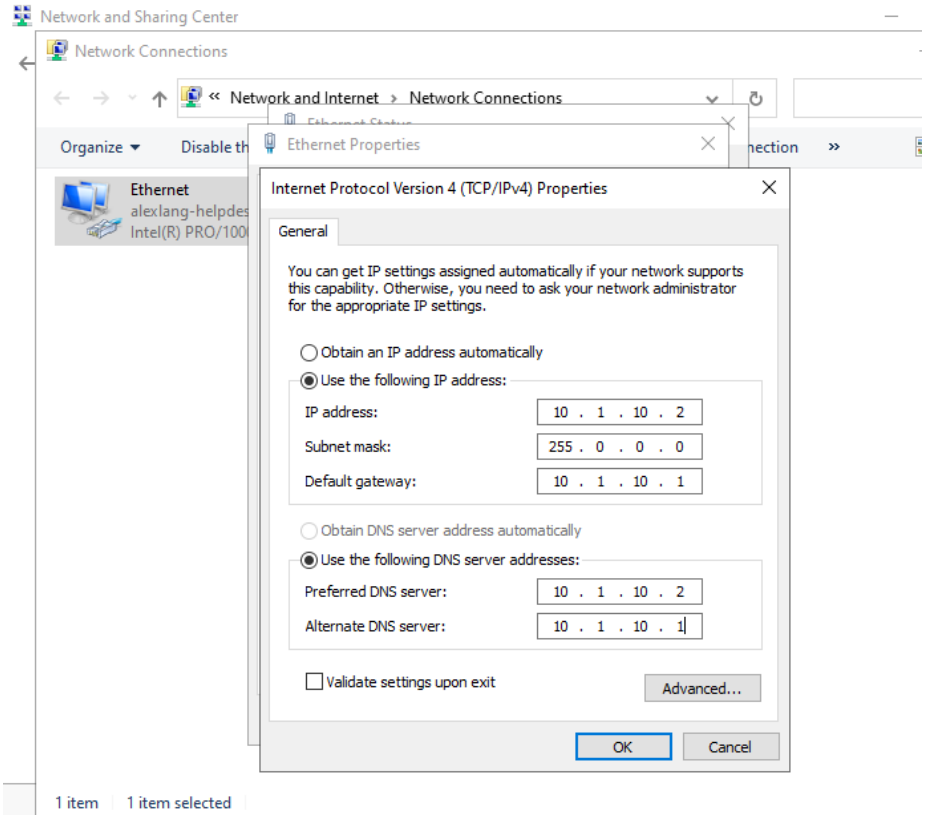
Went ahead and created another user account



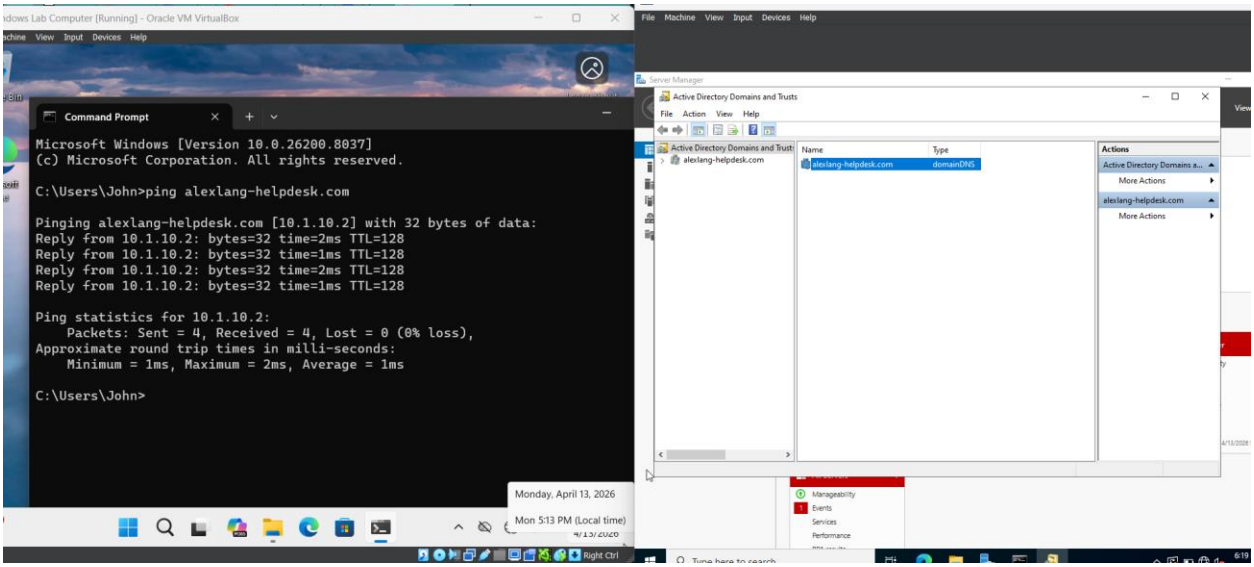
Reset password just to see how passwords get reset within ADs



Also updating network settings on Windows Server



Manually configuring IP addresses



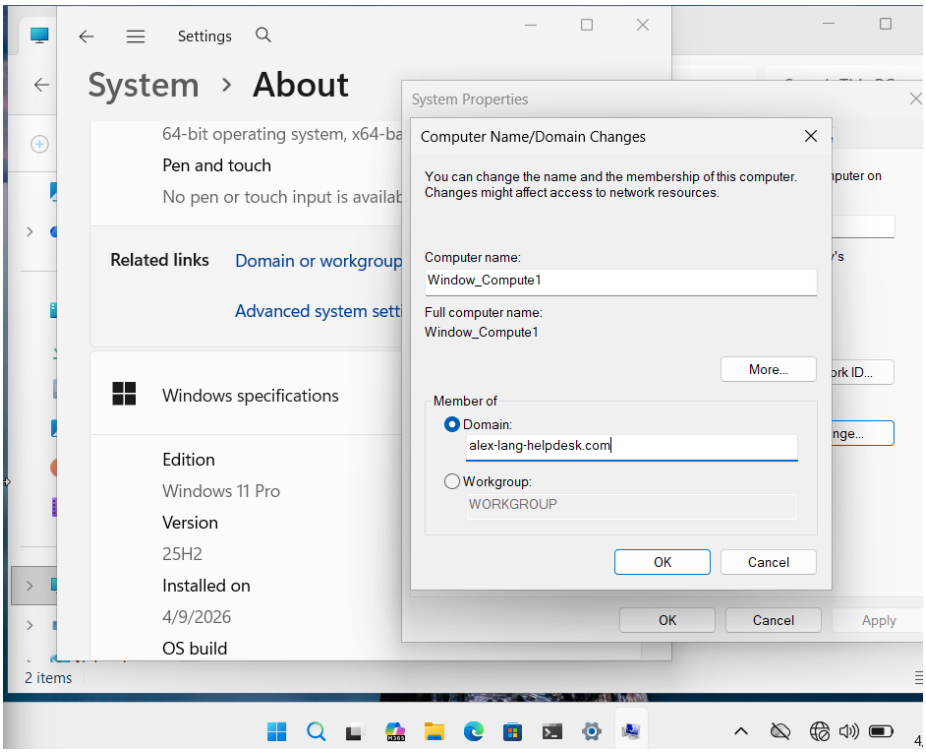
Using the ping command on one of the windows laptops to ensure that the domains IP address can be reached

```
Command Prompt

Examples:
    WHOAMI
    WHOAMI /UPN
    WHOAMI /FQDN
    WHOAMI /LOGONID
    WHOAMI /USER
    WHOAMI /USER /FO LIST
    WHOAMI /USER /FO CSV
    WHOAMI /GROUPS
    WHOAMI /GROUPS /FO CSV /NH
    WHOAMI /CLAIMS
    WHOAMI /CLAIMS /FO LIST
    WHOAMI /PRIV
    WHOAMI /PRIV /FO TABLE
    WHOAMI /USER /GROUPS
    WHOAMI /USER /GROUPS /CLAIMS /PRIV
    WHOAMI /ALL
    WHOAMI /ALL /FO LIST
    WHOAMI /ALL /FO CSV /NH
    WHOAMI /?

C:\Users\John>whoami /fqdn
ERROR: Unable to get Fully Qualified Distinguished Name (FQDN) as the current
       logged-on user is not a domain user.

C:\Users\John>
```

Logged into the computer as John which is a user I created that I thought I gave domain access too, but I guess I didn't

```
Windows IP Configuration

   Host Name . . . . . . . . . . . . : Window_Compute1
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
   Physical Address. . . . . . . . . : 08-00-27-7D-D8-86
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::7174:3313:7a8d:4213%9(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.1.10.3(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.0.0.0
   Default Gateway . . . . . . . . . : 10.1.10.1
   DHCPv6 IAID . . . . . . . . . . . : 101187623
   DHCPv6 Client DUID. . . . . . . . : 00-01-01-00-31-6A-14-68-08-00-27-7D-D8-8
   DNS Servers . . . . . . . . . . . : 10.1.10.2
                                       10.1.10.1
   NetBIOS over Tcpip. . . . . . . . : Enabled

C:\Users\John>ipconfig
```

Using the ipconfig /all command to ensure that the DNS server is pointing where I want it, which is it 10.1.10.2 and 10.1.10.1

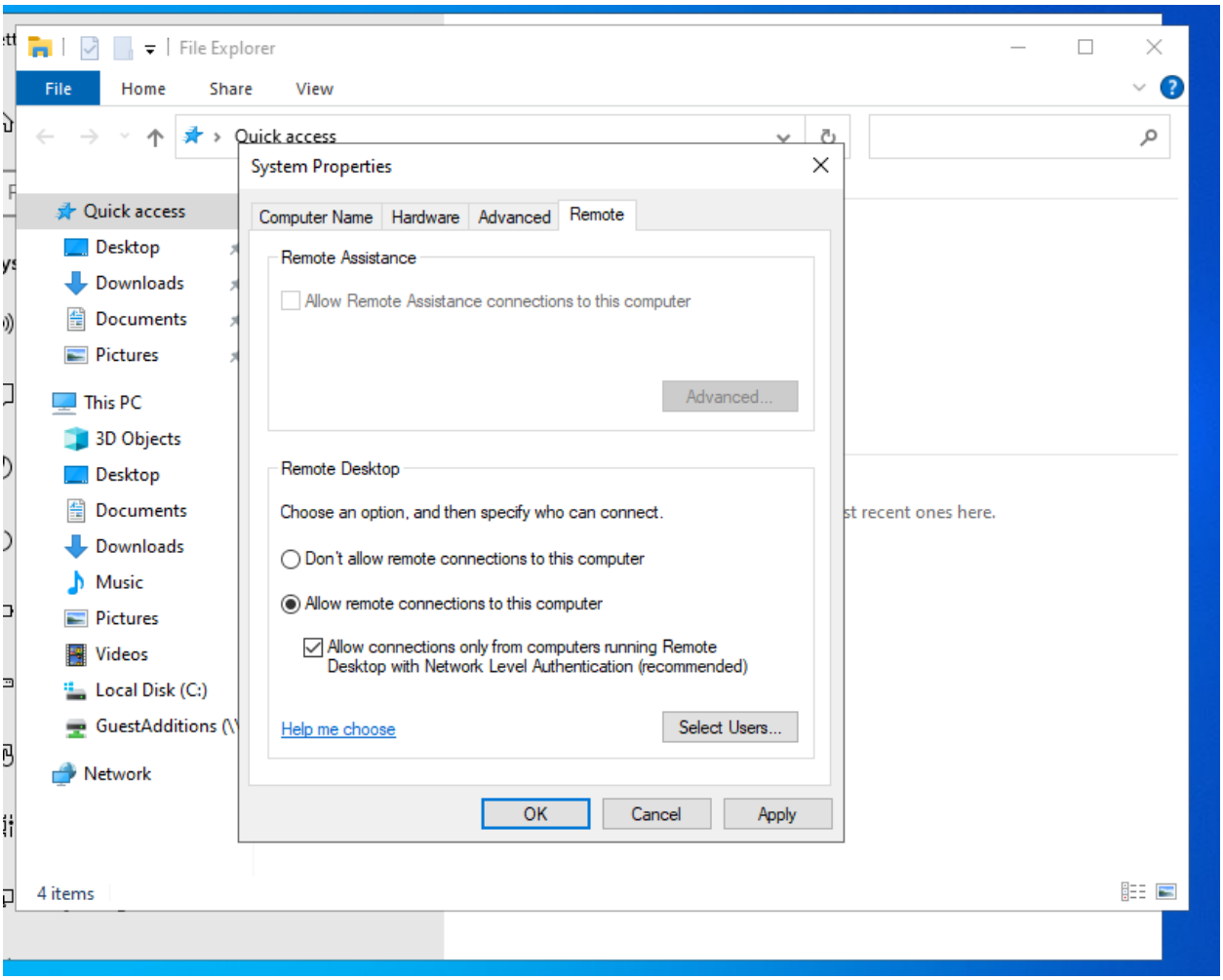


Went into properties->Domain or workgroup and then changed which domain this computer was apart of.

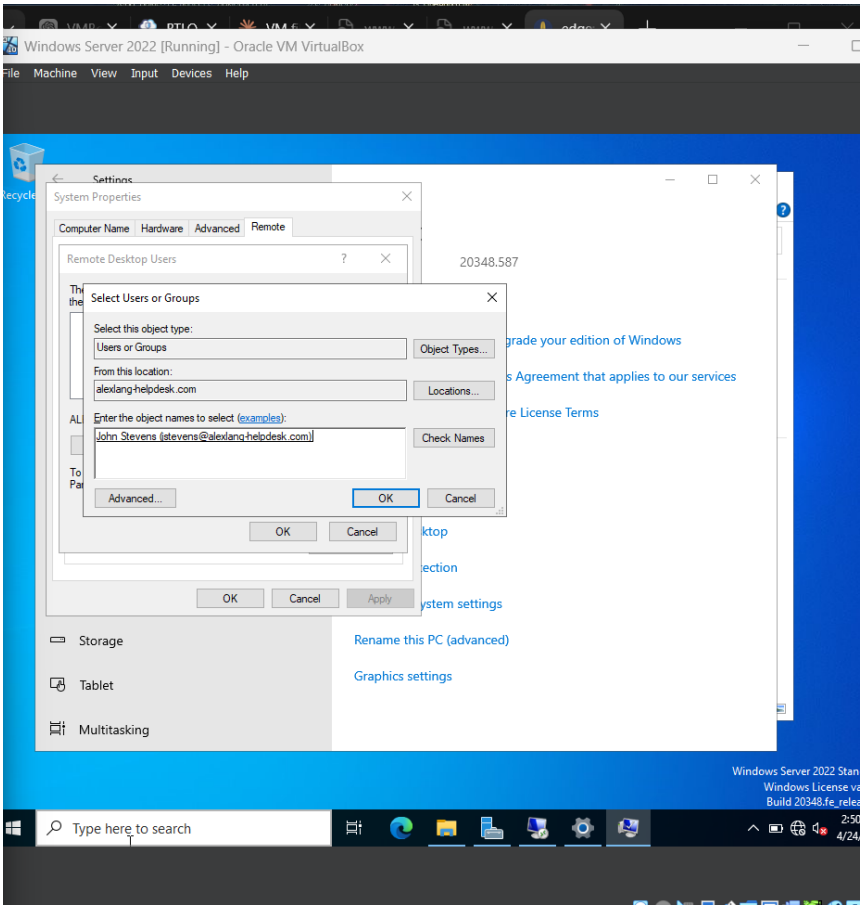
Observations/Problems that I had to troubleshoot through

- Had to troubleshoot John not being a domain access user (logged in as a local user and not a domain user)

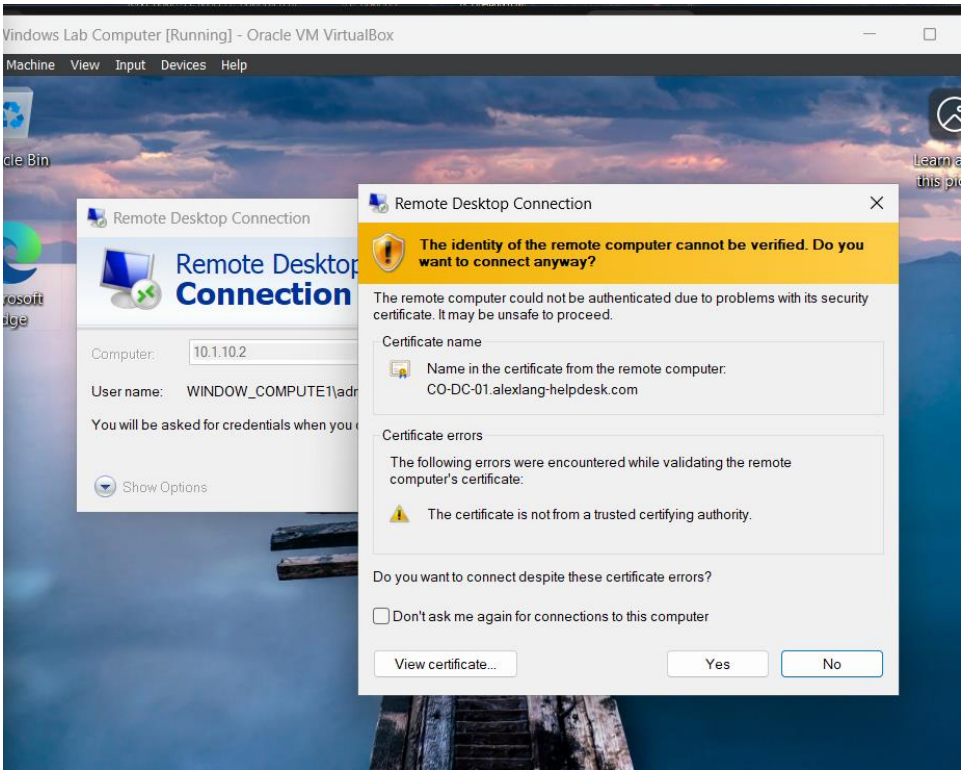
Day 7 Setting up RDP and Group Policy



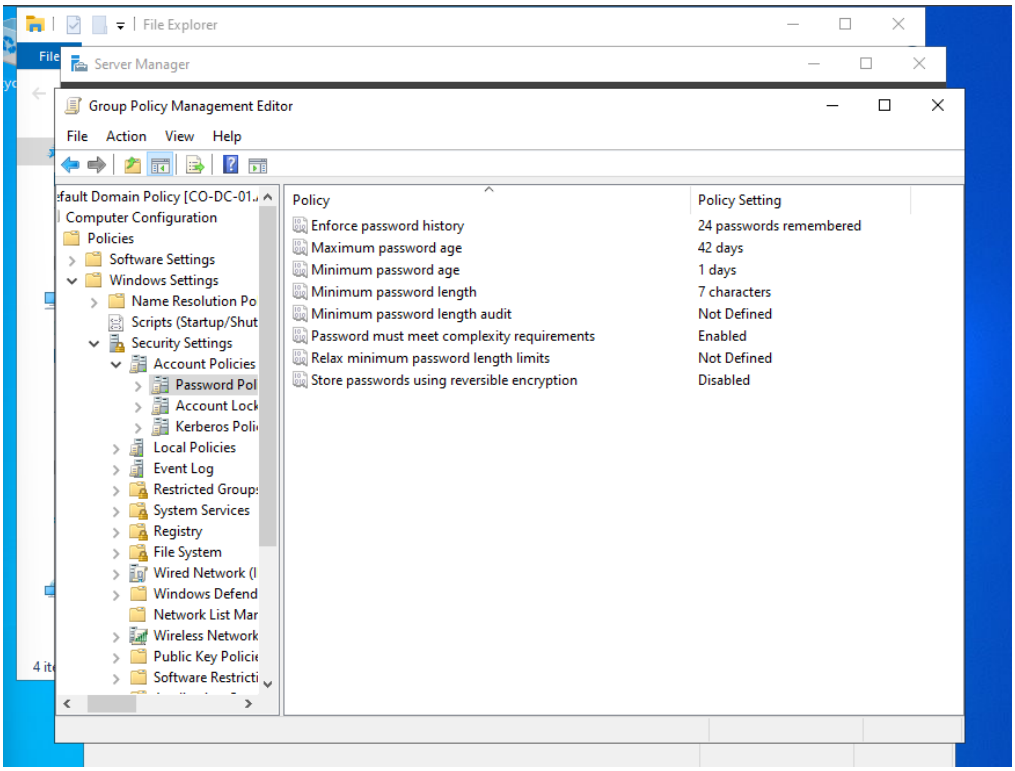
Setting up RDP by allowing remote connections inside of system properties



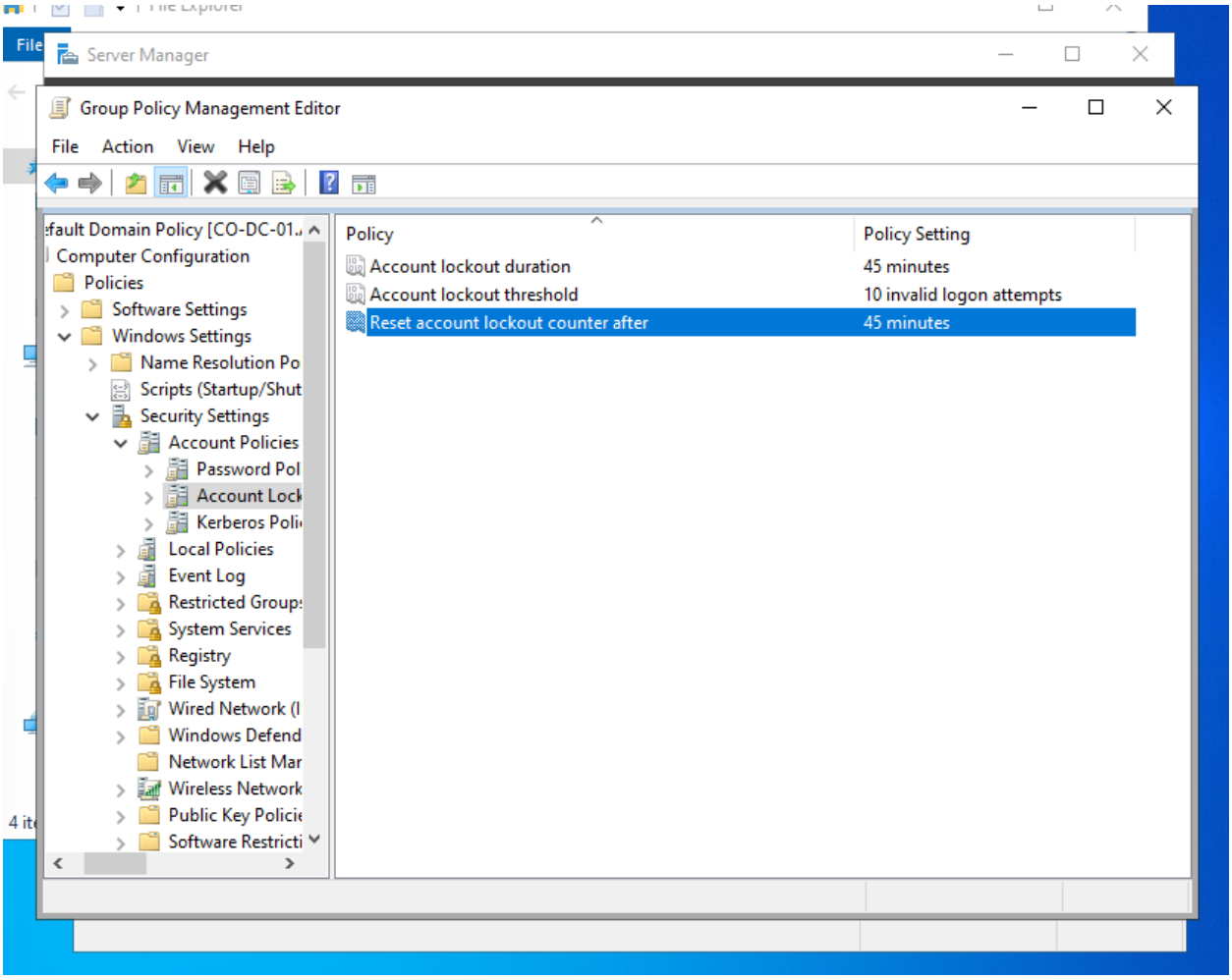
Adding a domain user to be able access the server via RDP



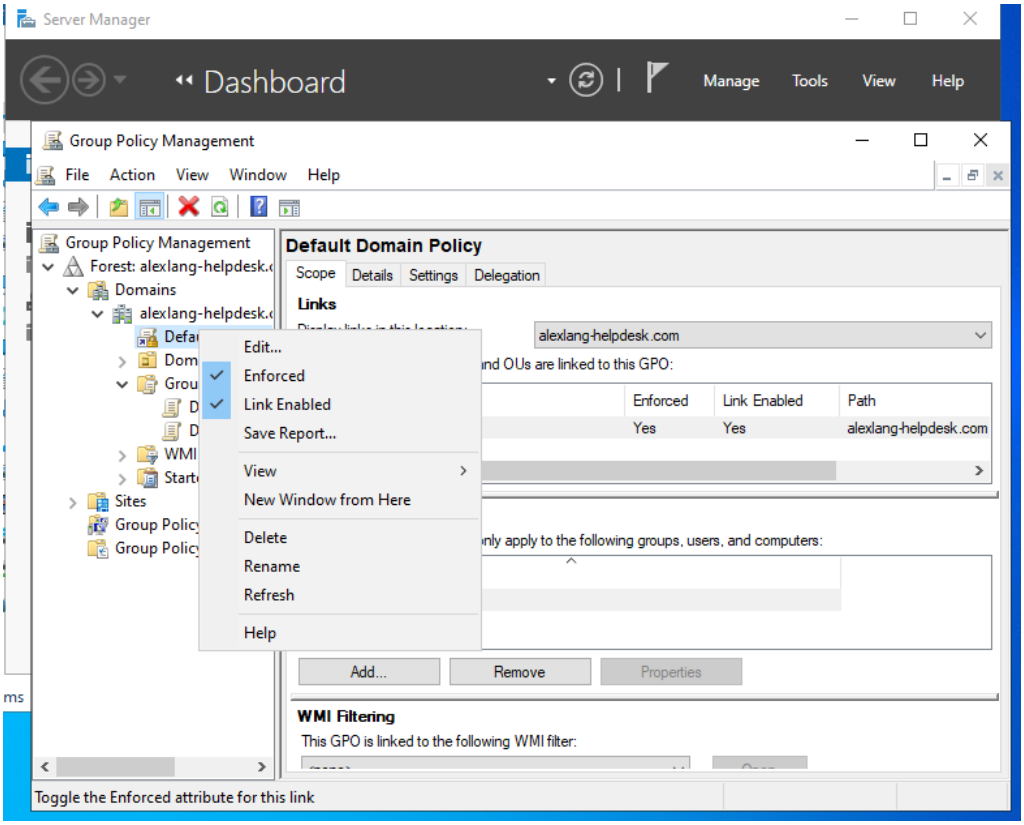
Sucessfully connected to the server now via RDP



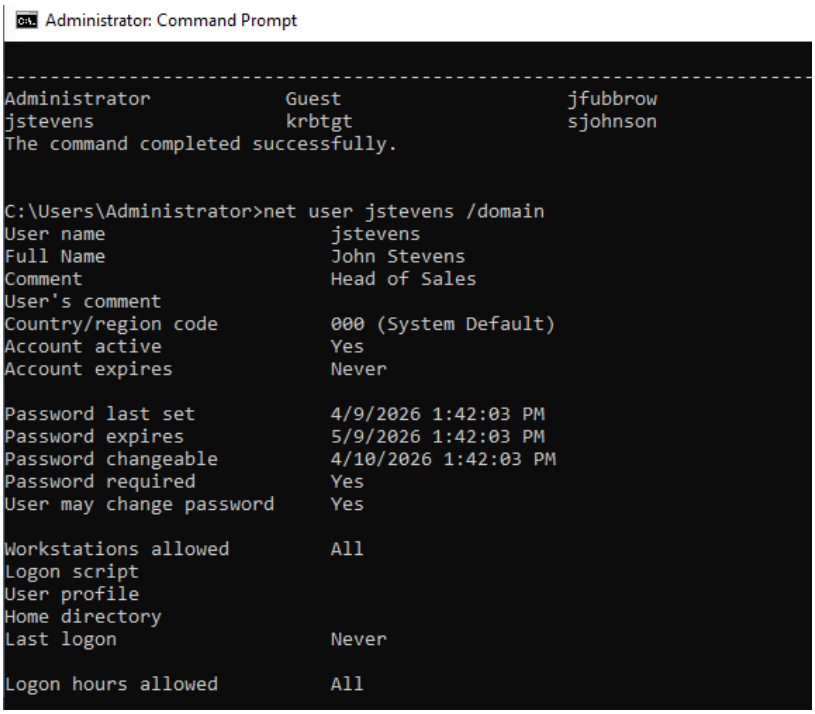
Edited group policy editor password settings so that passwords are to be reset every 30 days, and passwords must be at least 14 characters long



Also did some security hardening by increasing account lockout duraction, and longer account reset lockout time also within the account lockout settings



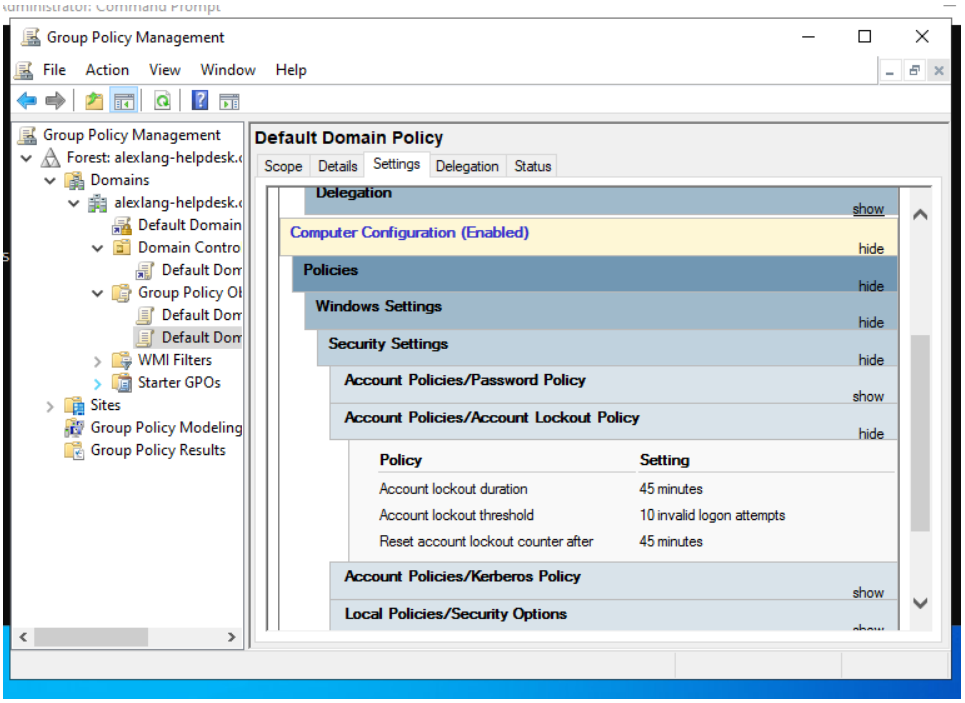
Enforcing the policy just created



After running net usre jstevens /domain, showing that the policy is correctly being enforced

```
COMPUTER SETTINGS
------------------
    CN=CO-DC-01,OU=Domain Controllers,DC=alexlang-helpdesk,DC=com
    Last time Group Policy was applied: 4/24/2026 at 3:24:39 PM
    Group Policy was applied from:      CO-DC-01.alexlang-helpdesk.com
    Group Policy slow link threshold:   500 kbps
    Domain Name:                        ALEXLANG-HELPDE
    Domain Type:                        Windows 2008 or later

    Applied Group Policy Objects
    ----------------------------
        Default Domain Policy
        Default Domain Controllers Policy

    The following GPOs were not applied because they were filtered out
    -------------------------------------------------------------------
        Local Group Policy
            Filtering:  Not Applied (Empty)

    The computer is a part of the following security groups
    -------------------------------------------------------
        BUILTIN\Administrators
        Everyone
        BUILTIN\Pre-Windows 2000 Compatible Access
        BUILTIN\Users
```

Viewing gp information from the command prompt after running gpresult /R

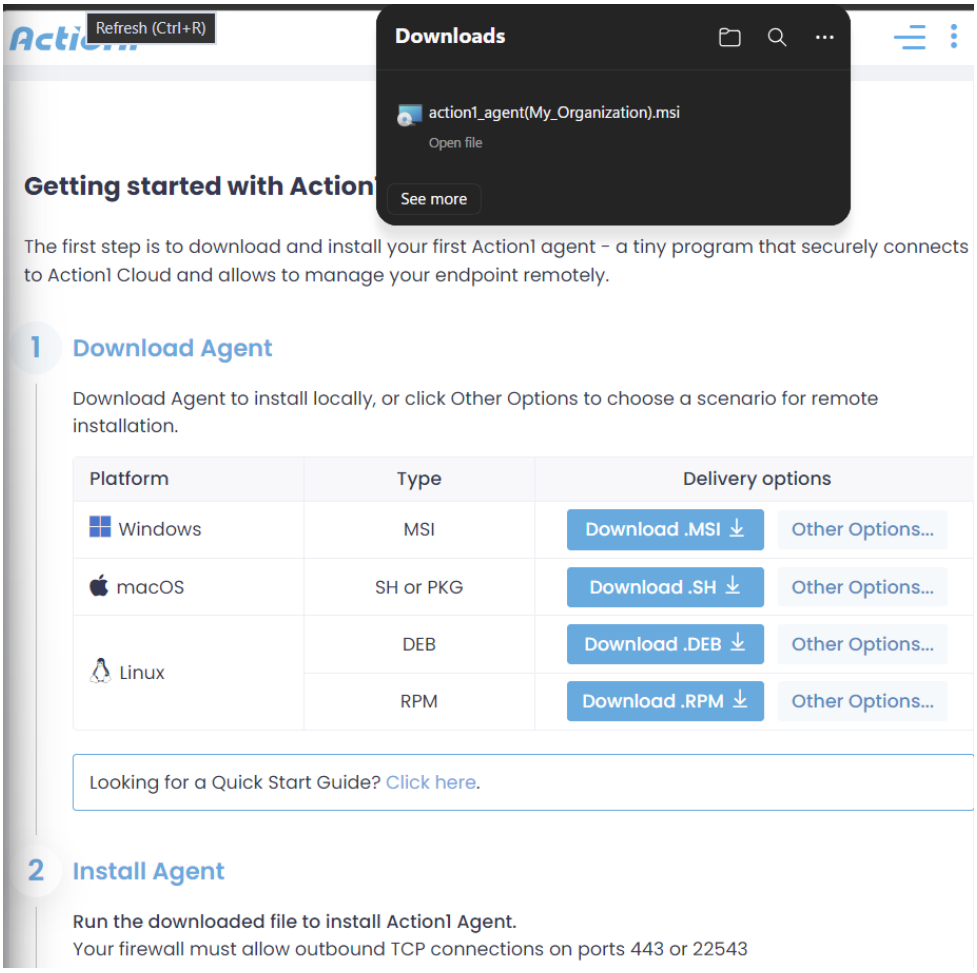


Also viewing the new updated policy within the group policy management tab

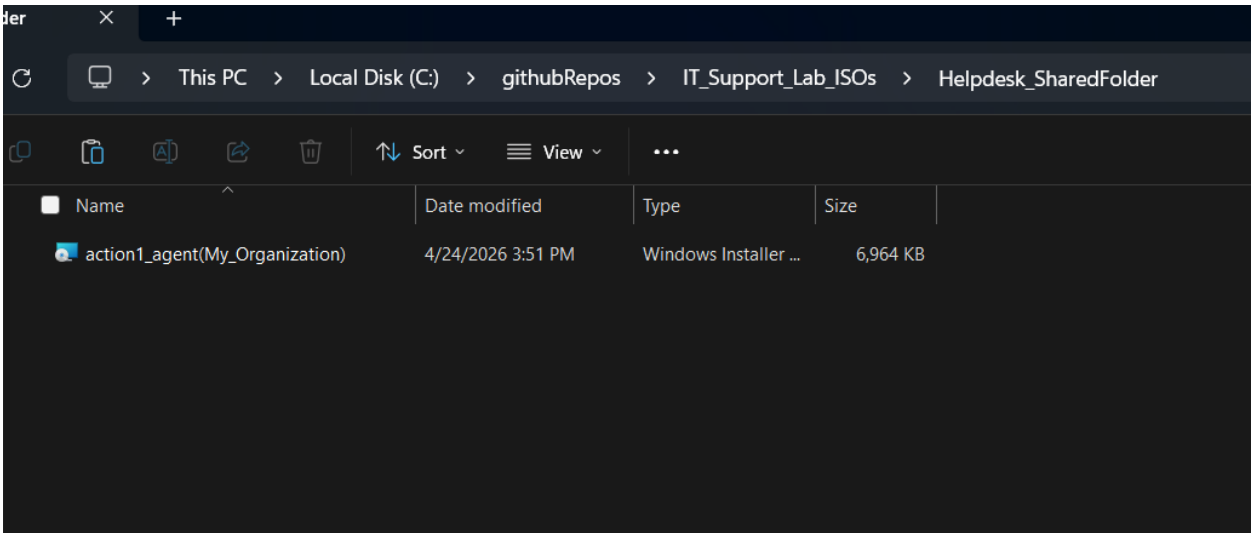
Observations/Problems that I had to troubleshoot through

- When I first edited the gp, I did not enforce it so the changes did not change(so had to go back and enforce it)

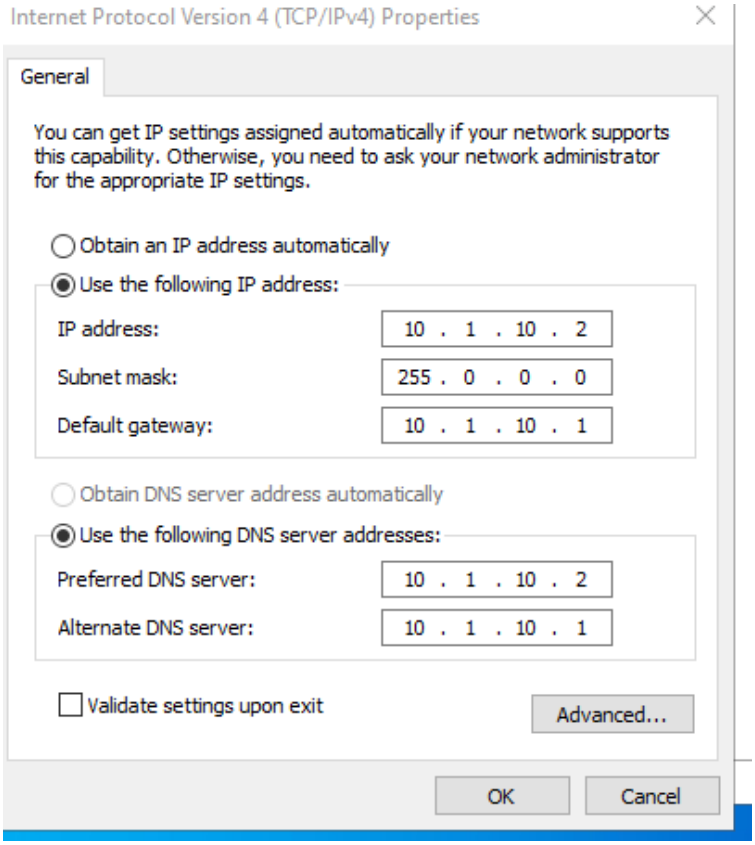
Day 8 Patch Management using Action 1, Reports, and Audit Documentation

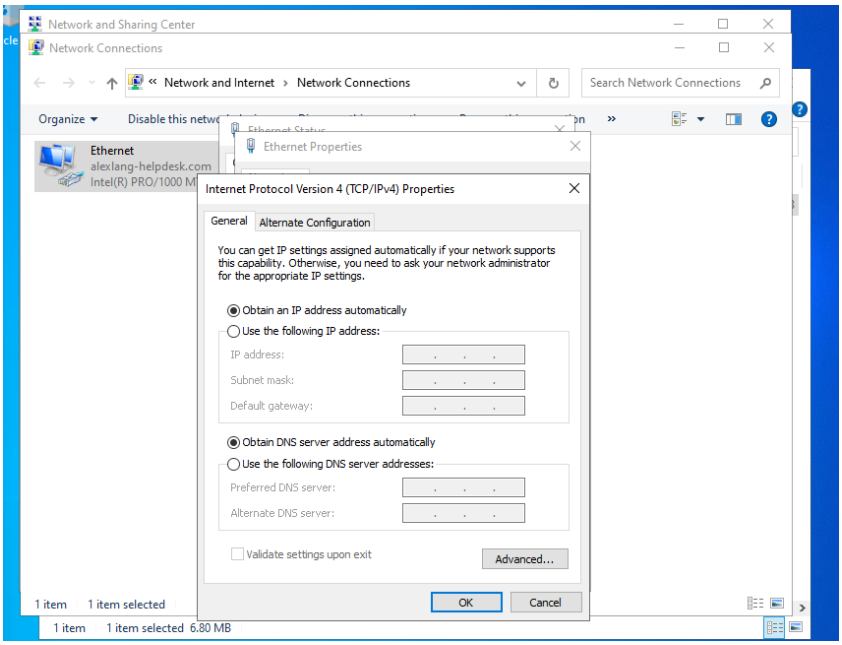


Went ahead and created an account with Action1, and downloaded the msi

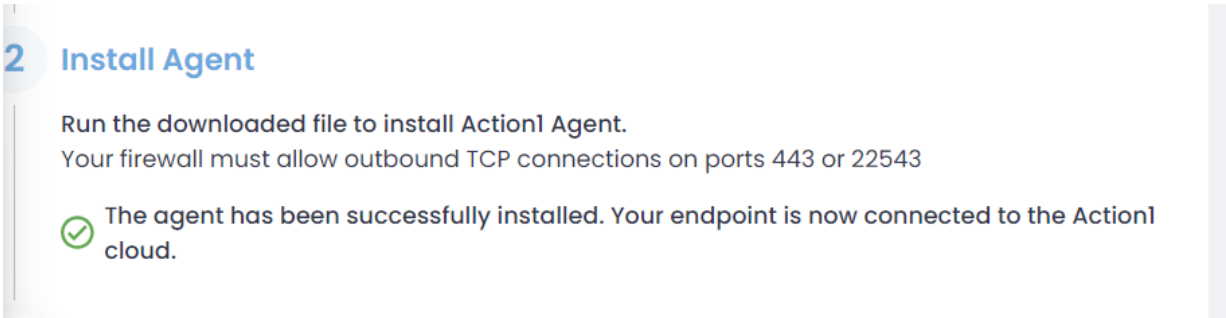


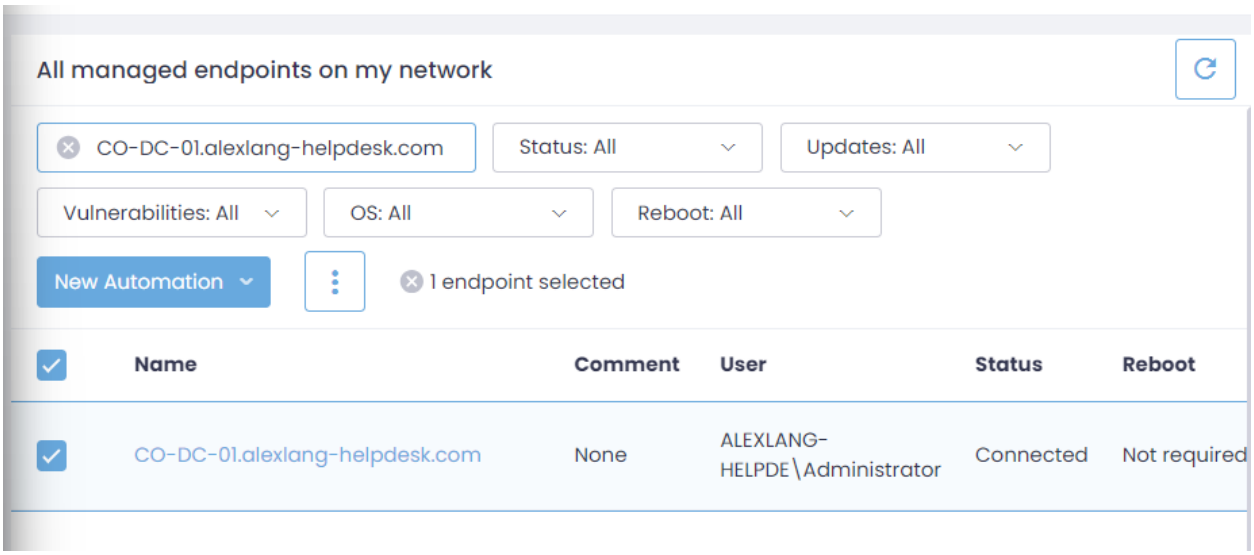
Created a shared folder on my home os, copied the msi file to the folder, and then added the shared folder to both



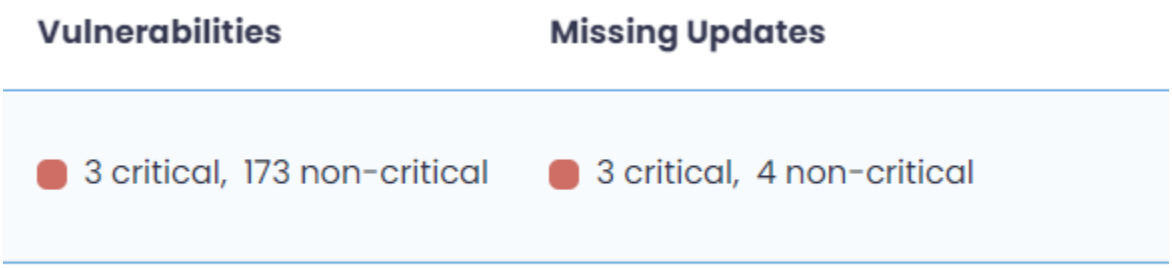


So currently the VM does not have internet, so I needed to temporaliy enable it but turning on DHCP, turning off the static IP address, and temporality making it a bridged adapter instead of a host only adapter. Will need to change back after this process is done.

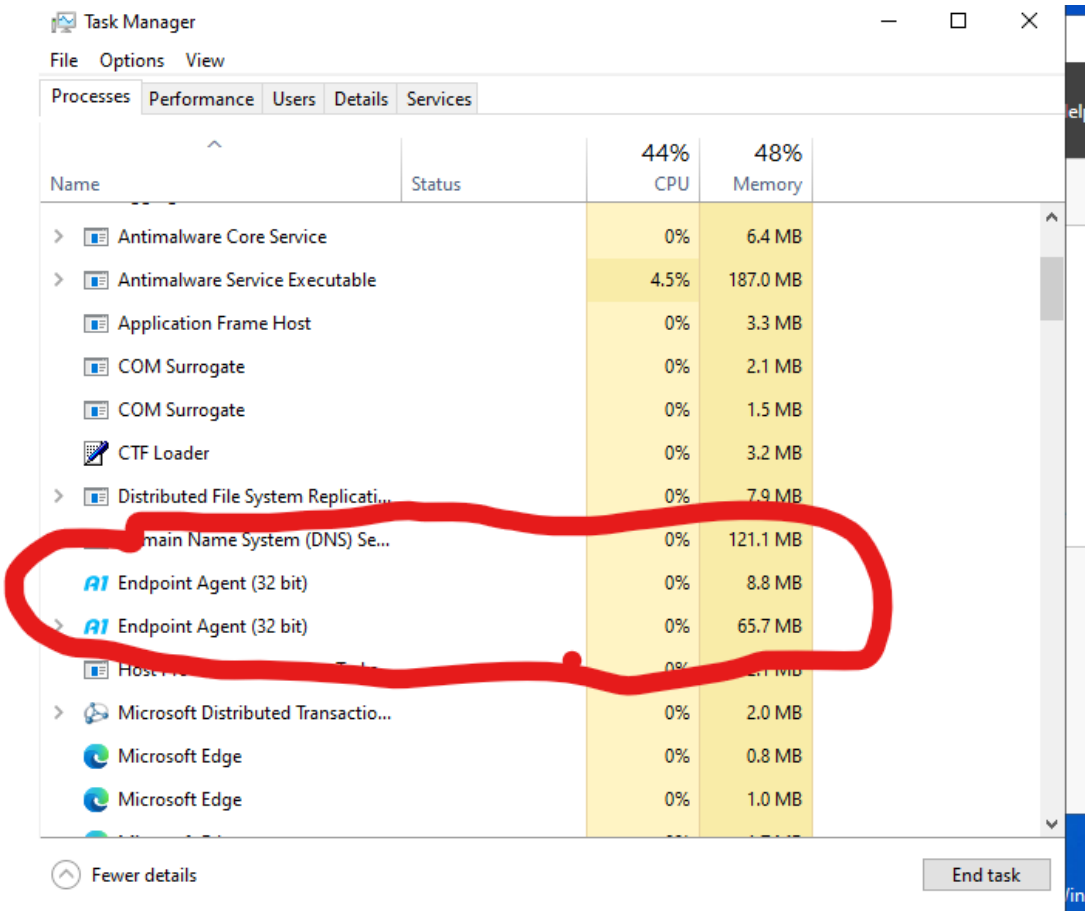




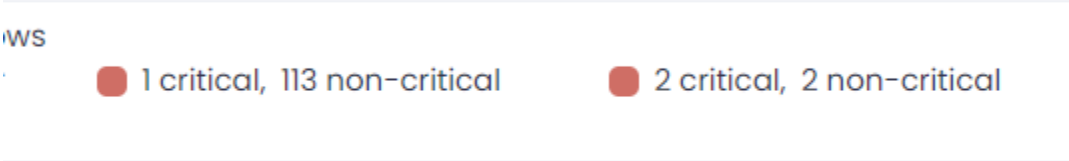
Agent now sucessfully configured



Right away, I noticed that the agent has 3 critial vunerabilities and 3 missing updates so need to be taken care of



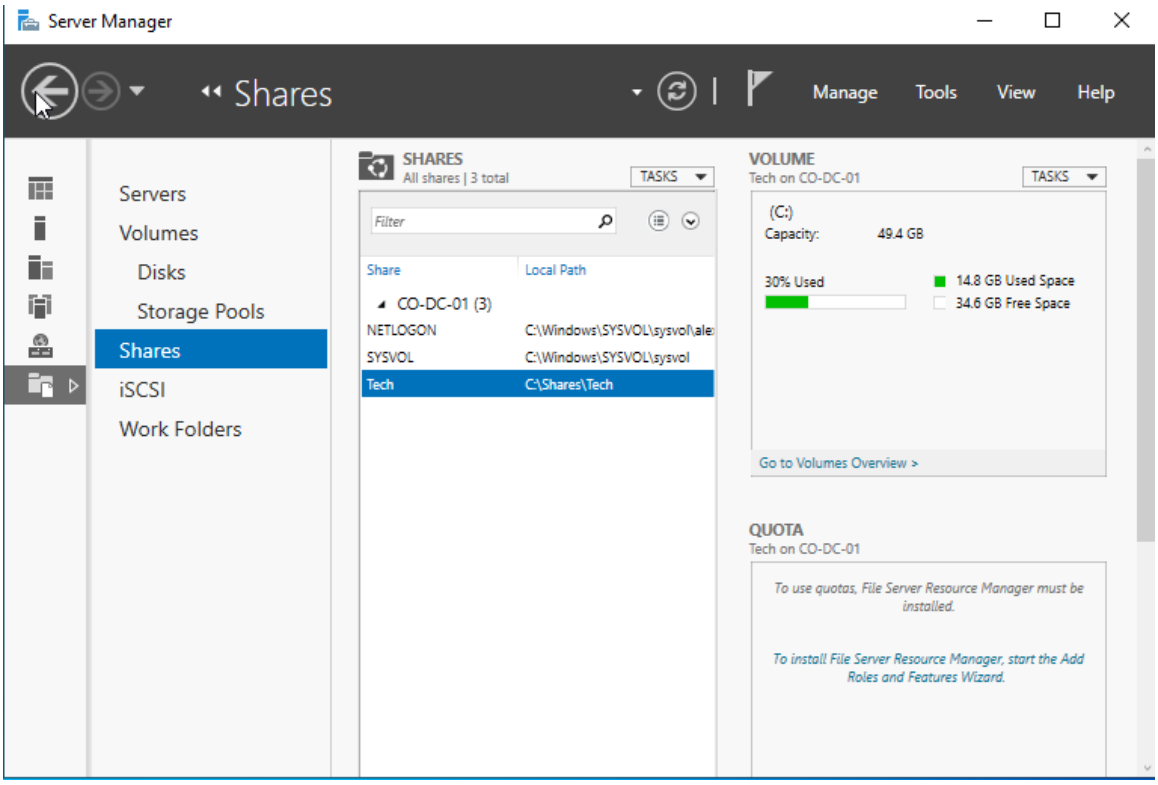
Agent successfully running/running inside of Server



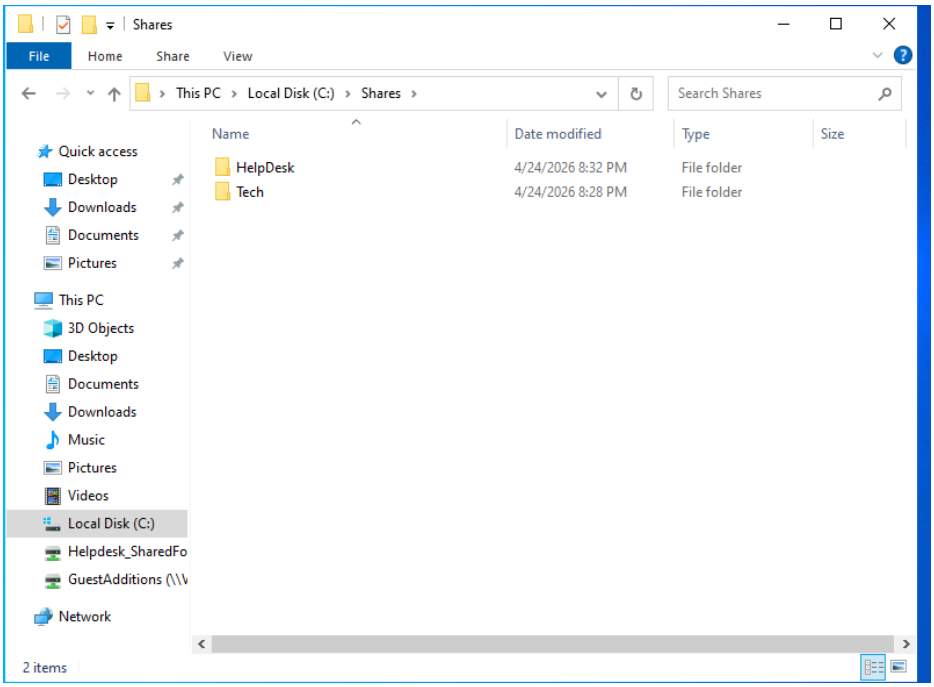
Ran a bunch of updates, and decreased the number of vulnerabilities

Observations/Problems that I had to troubleshoot through

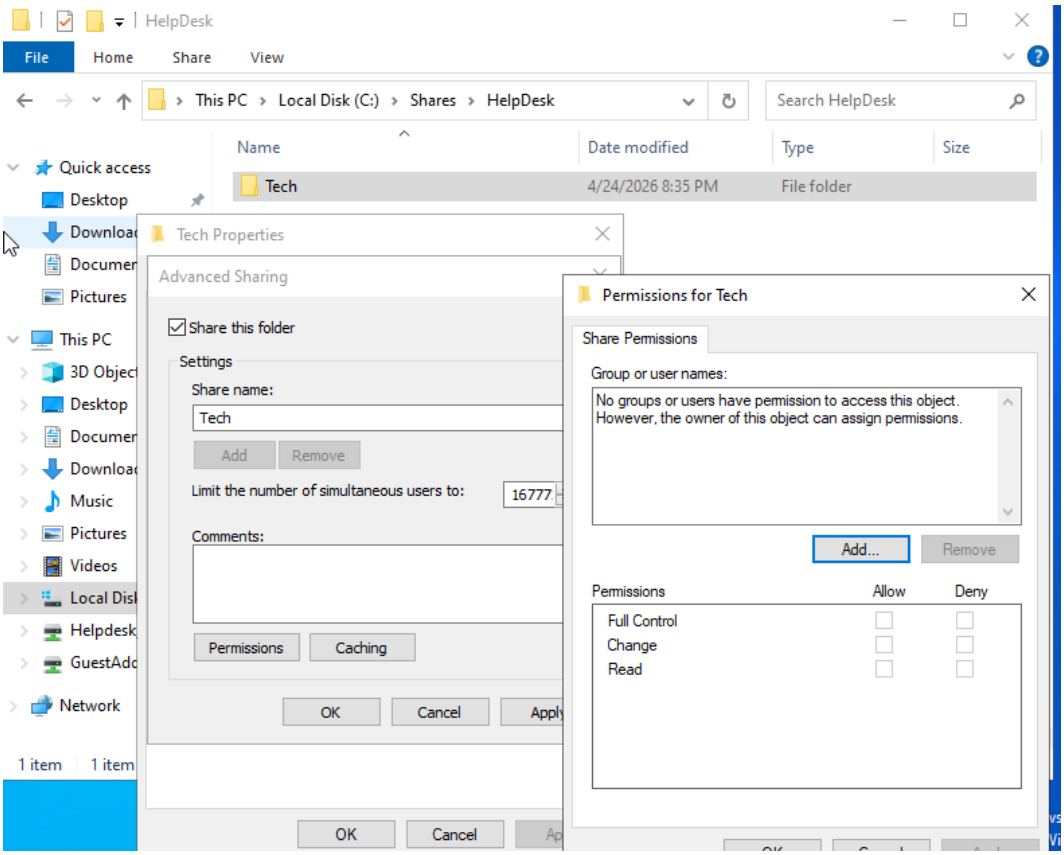
Day 9 Shared Files



Created a new share



Shared folder is now created, and shows up on the C drive



Since most accounts don't need access to this, went ahead and disabled access by default. I did add myself through since I'm an administrator.

Observations/Prolbems that I had to troubleshoot through

Day 10 Group Policy and Changing Users Settings

Observations/Prolbems that I had to troubleshoot through

Day 11

Observations/Prolbems that I had to troubleshoot through

Day 12

Observations/Prolbems that I had to troubleshoot through

Day 13

Observations/Prolbems that I had to troubleshoot through

Day 14

Observations/Prolbems that I had to troubleshoot through